

CHAPTER ONE

THEORY OF CONSUMER BEHAVIOR & UTILITY MAXIMIZATION

Dear student! After completing this chapter, you will be able to:

- Explain the concept of cardinal and ordinal utility approach;
- Identify the difference between the indifference curve and indifference map;
- Discuss the properties of indifference curve;
- Derive the budget line
- State and explain consumer and producer surplus;
- Derive the market demand;

1.1. Introduction to Theory of Consumer Behavior

The theory of demand starts with the examination of the behavior of the consumers. In our everyday life we behave in different ways while buying and consuming a good or service. The simple calculations and human reasoning we undertake while doing any transactions have been transformed into principles which guide us to attain satisfaction or equilibrium in economic sense.

 *Dear student! Can you explain the concept of consumer behavior in your own word?*

Consumer Behavior is meant how consumers decide on the basket of goods and services they consume or is the way consumers take decisions among various alternatives. Consumer behavior is best understood in three distinct steps:

Step 1: Consumer Preference

Specifically, we need a practical way to describe how people might prefer one good to another.

Step 2: Budget Constraint

In this step, we must take in to account for the fact that consumers face budget- constraint- they have limited incomes that limits the quantities of goods that they can buy.

Step 3: Consumer Choices

Given their preferences and limited incomes, what combinations of goods will buy to maximize their satisfaction? In other words, given their preferences and budget constraints (limited income), it is possible to determine how individual consumers choose how much of each good to

buy. The basic assumption is that consumers make this choice in a rational way – that they choose goods to maximize the satisfaction they can achieve, given the limited budget available to them.

1.2. Consumer Preference and Utility

 ***Dear distance learner! Can you explain the difference between consumer preference and utility?***

❖ **Consumer Preference**: we mean what consumers like and dislikes. The theory of consumer behavior begins with three basic assumptions regarding people's preferences for one market basket versus another. These assumptions hold for most people in most situations:

A) Completeness: Preferences are assumed to be ***complete***. In other words, consumers can compare and rank all possible baskets. Thus, for any two market baskets A and B, a consumer will prefer A to B ($A > B$), will prefer B to A ($B > A$), or will be indifferent between the two ($A = B$). (By "***indifferent***" we mean that a person will be equally satisfied with either basket.) Note that these preferences ignore costs. A consumer might prefer steak to hamburger but buy hamburger because it is cheaper. Therefore, it is hardly objectionable.

B) Transitivity: Preferences are ***transitive***. It means consumer preference should not be self-contradictory. In other words, Transitivity means that if a consumer prefers basket A to basket B ($A > B$) and basket B to basket C ($B > C$), then the consumer also prefers A to C ($A > C$). These preferences are normally regarded as necessary for consumer consistency.

C) More is better (Non satiation): consumers always prefer *more of any good to less*. In general, consumers always prefer larger quantity. (Assume goods are desirable; exclude undesirable goods.)

 ***Dear distance learner! Can you define the word "Utility" in your words?***

❖ **Utility**: Utility is defined as the power of a commodity or service to satisfy a human want. It is subjective concept and therefore varies from person to person because every person varies in their likes and dislikes. We should note that utility is the '***expected satisfaction***' whereas satisfaction implies '***realized satisfaction***'.

In the language of economics, the concept of ***utility*** refers to the numerical score representing the satisfaction that a consumer gets from a market basket. In other words, utility is a device used to

simplify the ranking of market baskets. E.g. If buying three copies of a textbook gives you more pleasure than buying one shirt, then it is said that the books give you more utility than the shirt.

 ***Dear distance learner have you ever hear the approaches economists used to measure utility?***

There are two approaches to measure utility theory (to the analysis of consumer behavior):

- 1) Cardinal utility analysis and
- 2) Ordinal utility analysis

1.2.1. Cardinal Utility Analysis

A utility analysis that describes by how much one market basket is preferred to another is called *cardinal utility analysis*. Unlike ordinal utility analysis, a cardinal utility analysis attaches to market baskets numerical values that cannot arbitrarily be doubled or tripled without altering the differences between values of various market baskets.

Dear distance learner! You should know that cardinal utility is the view that utility is quantitative or cardinally measurable feeling. It is used by Classical and Neo- classical economists.

Assumptions

- 1) ***Rationality***: It is assumed that the consumers are rational being in the sense that he/she satisfies his wants in the order of their merit/utility. The rational consumer has a powerful reasoning in evaluating which commodity can satisfy him most. They can identify what is good what is bad.
- 2) ***Limited money income***: the consumers have a limited money income to spend on the goods and services he/she chooses to consume.
- 3) ***Maximization of utility***: Every rational consumer intends to maximize his/her satisfaction from his/her given money income.
- 4) ***Utilities Cardinally measurable***: the Cardinalists assume that utility can be measured in absolute terms/numerical numbers; 1,2,3.../. They hoped that individuals' preferences could be easily quantified or measured in terms of basic units. The units used to measure utility are known as *utils*.
- 5) ***Utility is subject to the law of diminishing marginal utility***: As the consumer uses more and more units of a certain commodity, the extra increment of utility he derives from each extra

unit of the good declines. It shows the inverse relationship between marginal utility & its quantity.

- 6) **Constant marginal utility of money (MUm)**: the MU of money remains constant whatever the level of consumer's income and each unit of money has utility equal to one unit (MUm=1).
- 7) **Utility is additive**: the additive of the utility can be expressed through the utility function. If $U = F(x_1, x_2, x_3, \dots, x_n)$, then the sum of utilities that individual can derive from consuming various units of commodities as: $TU_n = U_1(x_1) + U_2(x_2) + U_3(x_3) + \dots + U_n(x_n)$.

1.2.1.1. Total Utility (TU) and Marginal Utility (MU)

 ***Dear distance learner! Can you define the term “Total Utility” and “Marginal Utility” in your words?***

A) Total Utility (TU): The overall satisfaction enjoyed from consuming any given quantity of a good is called total utility. TU is the sum of the utility derived from all the units consumed of a commodity.

B) Marginal Utility (MU): The extra satisfaction a person receives over a given period by consuming one more unit of a good is called marginal utility. Thus, MU measures the extra or additional benefit that results from increasing the consumption of a single commodity by one more unit, holding the consumption of other commodities being equal.

Alternatively, we can define MU is the change in total utility resulting from the consumption of one extra unit of the product.

i.e. $MU = \frac{\Delta TU}{\Delta Q}$ (MU is the slope of the TU curve) Where, ΔTU and ΔQ are the changes in total utility and quantities (units) consumed respectively.

$$\text{Or } MU_n = \frac{TU_n - TU_{n-1}}{Q_n - Q_{n-1}} \quad \text{or} \quad MU_{n+1} = \frac{TU_{n+1} - TU_n}{Q_{n+1} - Q_n}$$

☞ For $\Delta Q \rightarrow 0$, $MU = \frac{\partial TU}{\partial Q}$

Example:

Table 1.1 Total utility and Marginal utility

Quantity(Q _x)	1	2	3	4	5	6	7	8	9
TU	12	22	30	36	40	40	39	37	32
MU	12	10	8	6	4	0	-1	-2	-5

Dear distance learner! The above table shows the consumer maximizes utility at zero MU when the TU of “X” is 40 units. At zero MU, the consumer reaches **saturation point**. Beyond this point, the MU becomes negative and the rational consumer will not cross the limit of consumption beyond 6 units of the commodity.

Dear distance learner! From the above concept we can build the Relationship between TU and MU as follows:

- ☞ When TU increases at an increasing rate, MU also increases (special case).
- ☞ When TU increases but at a decreasing rate, MU declines but positive. In other words, MU starts to continuously decline when the TU start to increasing at a decreasing rate.
- ☞ When TU reaches a maximum (constant level), MU is zero and this stage is a **saturation point**.
- ☞ When TU starts to fall, MU declines and it is negative, i.e. the consumers are dissatisfied.

1.2.1.2. The Law of Diminishing Marginal Utility (LDMU)

 **Dear distance learner! Do know how does MU of the commodity change when the quantity consumed of the commodity increases?**

To answer the question we need to study the basic law of economics known as the **law of diminishing marginal utility**. The law states that, as more and more of any single commodity is consumed during a specified time period, successive units usually yield smaller and smaller increases in satisfaction. The additional satisfaction attained from additional unit of a good is measured by the good’s MU than previous units. This principle is called the LDMU.

In other words, the law states that as the quantity of commodity consumed increases per unit of time, the utility derived from the successive units goes to decline provided the consumption of all other goods remains constant. For example, take the case of consumption of Banana. The first

banana he consumes brings him considerable amount of satisfaction. However, taking more and more bananas would mean deriving less and less satisfaction from each extra unit consumed. In fact, the rational consumer may stop eating after some amount of bananas which implies $MU=0$ (i.e. *Saturation point*).

1.2.1.3. Consumer Equilibrium Under Cardinal Utility Approach

A consumer is said to be at optimum/equilibrium when he/she maximizes utility subject to income constraint. The utility maximizing consumer reaches his equilibrium points when he allocates his money in such a way that the last birr spent on each commodity yields the same utility and his money income available for the purchase of the goods has been exhausted.

Consider $U = f(x)$, price of X, P_X .

The consumer can either buy an additional unit of X or keep the income, which was to be spent on that additional unit of X unspent. Both give some satisfaction to the consumer. To determine the optimum of the consumer will need to compare the MU_x and P_X .

$MU_x \Rightarrow$ which the consumer will have to pay.

$P_x \Rightarrow$ What the consumer has to pay.

$MU_x > P_x \Rightarrow$ the consumer can raise his/her satisfaction by purchasing more units of good of X.

$MU_x < P_x \Rightarrow$ The consumer can raise his level of satisfaction by reducing consumption of x.

$MU_x = P_x \Rightarrow$ Optimum of the consumer (TU is maximized).

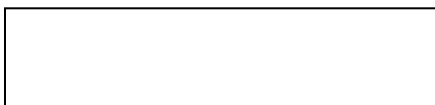
When the utility function of the consumer contains more than one good; $U = f(x_1, x_2, \dots, x_n)$. The optimum of the consumer receives the following conditions to be fulfilled.

$$MU_{x1} = P_{x1} \Rightarrow \frac{MU_{x1}}{P_{x1}} = 1$$

$$MU_{x2} = P_{x2} \Rightarrow \frac{MU_{x2}}{P_{x2}} = 1$$

$$\frac{MU_{x1}}{P_{x1}} = 1 = \frac{MU_{x2}}{P_{x2}} \xrightarrow{\frac{MU_{x1}}{P_{x1}} \rightarrow \frac{MU_{x2}}{P_{x2}}} \frac{MU_{x2}}{P_{x2}}$$

Therefore, the condition is



$$\frac{MU_{x_1}}{Px_1} = \frac{MU_{x_2}}{Px_2} = \dots = \frac{MU_{x_n}}{Px_n} \quad \rightarrow \text{This is known as the *law of Equi-MU*.$$

$\frac{MU_{x_1}}{Px_1}$ is the satisfaction the consumer derives by spending one unit of money on x_1 . Spending one unit of money should result in the same satisfaction regardless of the good it is spent on x_1 . If spending one unit of money results in higher satisfaction when it is spent on x_1 compared to other goods. The consumer can raise his satisfaction by spending more on x_1 and less on other unit the above condition is fulfilled.

1.2.1.4. Derivation of Demand Curve

Dear distance learner! The derivation of demand is based on the axiom of diminishing marginal utility. The marginal utility of commodity x may be depicted by a line with a negative slope (fig. 1.2). Geometrically the marginal utility of x is the slope of the total utility function $U=f(q_x)$. The total utility increases, but at a decreasing rate, up to quantity x and then starts declining (fig. 1.1).

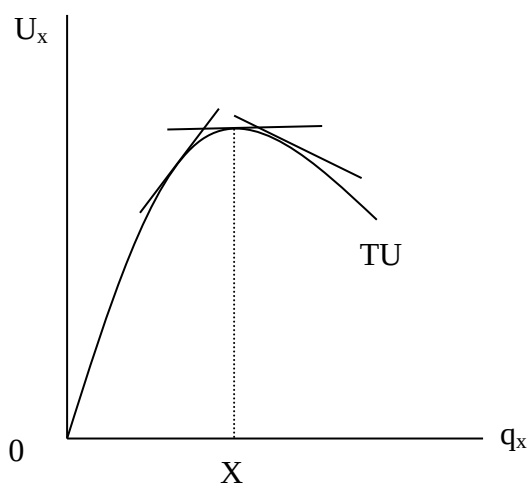


Fig. 1.1

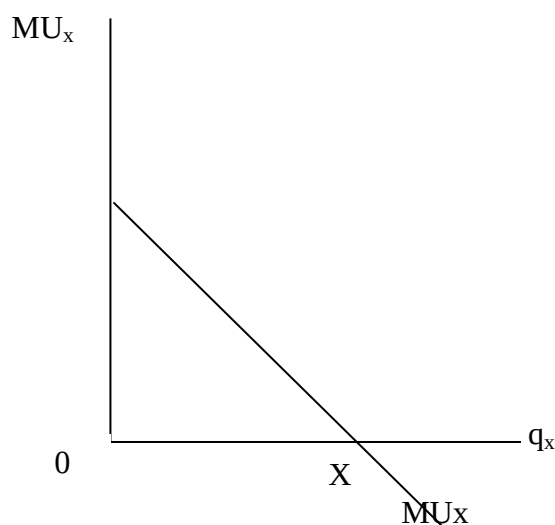


Fig. 1.2

Accordingly the marginal utility of x declines continuously; becomes negative beyond quantity x . If the marginal utility is measured in monetary units, the demand curve for x is identical to the positive segment of the marginal utility curve above the x -axis. At x_1 the marginal utility is MU_1 (fig. 1.3). This is equal to P_1 , by definition. Hence, at P_1 the consumer demands x_1 quantity (fig. 1.4).

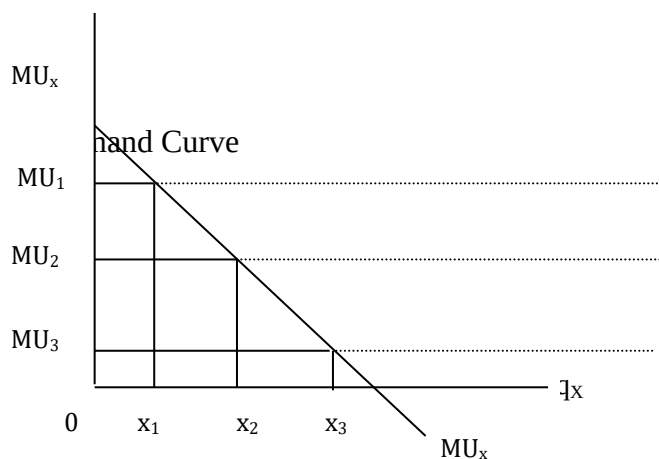


Fig. 1.3

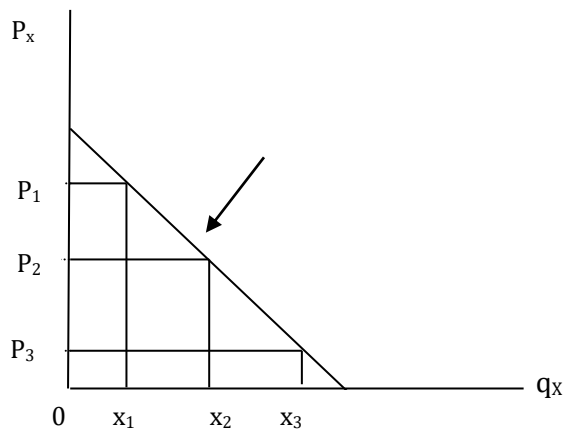


Fig. 1.4

It is clear that when the price of the commodity falls, the consumers buy more of the good so as to equate the MU to the lower price. At x_2 the marginal utility is MU_2 , which is equal to P_2 . Hence, at P_2 the consumer will buy x_2 , and so on. The negative section of MU curve does not form part of the demand curve because a rational consumer doesn't go beyond the zero level of MU, since negative quantities do not make sense in economics.

1.2.2. Ordinal Utility Approach

A utility analysis that generates ranking in order of most to least preferred of market baskets is called an *ordinal utility analysis*. In fact, numerical values are arbitrary, interpersonal comparisons of utility are impossible.

The only property of a utility assignment that is important is how it orders the bundles of goods. The magnitude of the utility function is only important insofar as it ranks the different consumption bundles; the size of the utility difference between any two consumption bundles does not matter. Because of this emphasis on ordering bundles of goods, this kind of utility is referred to as **ordinal utility**. It is also known as an **Indifference Curve Approach**. Here utility is not a quantitative or a numerical value. In other words, utility is not measurable rather it is comparable.

Assumptions

- 1) **Rationality:** The rational consumer has a powerful reasoning in evaluating which commodity can satisfy him most. They can identify what is good what is bad.

- 2) **Utility is Ordinal:** Utility is only ordinally measurable that is the consumer is able to express only the rank of his preference.
- 3) **Transitivity and Consistency of choice:**
 - ✓ **Transitivity:** If $A > B$, and $B > C$, then $A > C$.
 - ✓ **Consistency:** If $A > B$ in one period, should not be $B > A$, or $A = B$ for another period.
- 4) The total utility of the consumer depends on the quantities of the commodities consumed.
- 5) **Diminishing Marginal Rate of Substitution:** This rate is given by $\frac{\Delta Y}{\Delta X}$. The assumption is that $\frac{\Delta Y}{\Delta X}$ goes on decreasing, when consumer continuous to substitute X for Y.
- 6) **No Satiation:** a consumer always prefers a larger quantity of all the goods. In other words, this is the 'more is better than less' assumption.

❓ **Dear distance learner! Compare and contrast the assumptions of “Cardinal Approach” and “Ordinal Approach”?**

1.2.2.1. The Indifference Curve

Dear distance learner! It is an ordinal approach adopted for utility analysis. The Indifference curve(IC) is a locus of different combinations of two or more goods which yield the same level of satisfaction to the consumer. It is also called an **Iso-utility curve**. This is so, because moving along the same indifference curve leaves the consumer neither better nor worse-off. In other words, the consumer is indifferent between the different consumption bundles. Graphically,

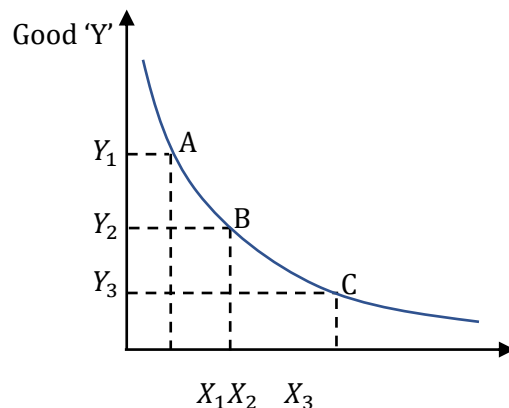


Fig. 1.5: The Indifference Curve

☞ All points on the indifference curve give the same level of satisfaction.

☞ The area between X and Y axis is known as **Indifference plane** or **Commodity Space**.

❖ **An Indifference Map:** is a set of indifference curves that describes a person's preferences. Each indifference curve representing different level of satisfaction for the consumers. But we should note that a higher indifference curve drawn up to the right of another on an indifference map yields a higher level of satisfaction to the consumer. For example, in $IC_3 > IC_2 > IC_1$ as shown in fig. 1.6 in the below diagram. In contrast, a lower indifference curve is likely to yield a low level of satisfaction to the respective consumer.

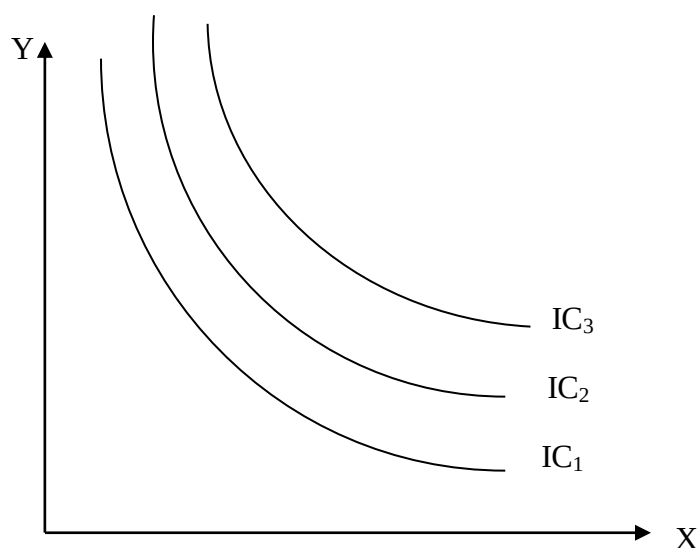


Fig 1.6. Indifference Map

1.2.2.2. The Marginal Rate of Substitution(MRS)

Dear distance learner! To quantify the amount of one good that a consumer will give up to obtain more of another, we use a measure called the *marginal rate of substitution* (MRS). The MRS is the rate of which one commodity can be substituted for another, the level of satisfaction remaining the same.

The *MRS* of x for y is the amount of y that a person is willing to give up to obtain one additional unit of x . Suppose, for instance, the MRS is 3. This means that the consumer will give up 3 units of y to obtain 1 additional unit of x . If the MRS is $1/2$, the consumer is willing to give up only $1/2$ unit of y so that the consumer maintains the same level of satisfaction. Thus, the MRS measures the value that the individual places on 1 extra unit of one good in terms of another.

The MRS_{xy} can be expressed by the slope of the indifference curve.

$$MRS_{xy} = \frac{\text{Number of units of } y \text{ give up}}{\text{Number of units of } x \text{ gained}} = \frac{\Delta y}{\Delta x}$$

Here, the consumption for good X increase and the consumption for good Y decrease.

Therefore, MRS is the slope of the indifference curve.

? Dear student! Do you know the difference between MRS_{xy} and MRS_{yx} ? Explain it?

Here it is important to note that MRS_{xy} is different from MRS_{yx} because $MRS_{xy} = \frac{\Delta y}{\Delta x}$, but $MRS_{yx} = \frac{\Delta x}{\Delta y}$. And You should note that the concept of MU is implicit in the definition of the MRS .

Proof

Given $U = f(x, y)$

$$du = \frac{\partial u}{\partial x} \cdot dx + \frac{\partial u}{\partial y} \cdot dy \dots \dots \dots \text{total differential}$$

$$0 = MU_x \cdot dx + MU_y \cdot dy$$

B/c ΔU in an indifference curve is zero since consumers are in the same level of satisfaction along the same indifference curve.

$$\text{since, } MU_x = \frac{\Delta u}{\Delta x} = \frac{\partial u}{\partial x} \text{ and } MU_y = \frac{\Delta u}{\Delta y} = \frac{\partial u}{\partial y}$$

$$\frac{MU_x}{MU_y} = -\frac{dy}{dx} = MRS_{xy}$$

$$\text{In the same manner, } MRS_{yx} = \frac{MU_y}{MU_x} = -\frac{dx}{dy}$$

☞ As we move from left to right on an indifference curve, the MRS_{xy} decreases in absolute values, this is referred to as the decreasing MRS ($LDMRS$).- It is not true for MRS_{yx} .

☞ As consumers receives more and more of a particular good, its value in terms of other goods declines.

- ❖ **Law of Diminishing Marginal Rate of Substitution:** The law states as more and more of one good are consumed, we can expect that a consumer will prefer to give up fewer and fewer units of a second good to get additional units of the first one. As we moved down the indifference curve in figure A page 9 and consumption of good X increases, the additional satisfaction that a consumer gets from still more food will diminish. Thus, he will give less and less good Y to obtain additional unit of good X. Here it is important to note that an indifference curve is convex if the *MRS* diminishing along the curve.

1.2.2.3. Characteristics of Indifference curve

Indifference curves have certain characteristics that reflect assumptions about consumer behavior. In fact, one of the major uses of indifference curves is to examine the kinds of consumer behavior implied by different preferences, prices, and incomes.

A) Indifference curves are normally negatively sloped. It implies that

- ✓ The two commodities can be substituted each other.
- ✓ That is quantities of one commodity decreases the quantity of the other commodity is increases if the consumer has to stay at the same level of satisfaction.

B) ICs are convex (bowed inward) to the origin. It implies that

- ✓ The two commodities are imperfect substitutes for each other. In other words, convexity of IC enables the consumer to buy more of good X and less good Y and yet the consumer remains on the same level of satisfaction.
- ✓ It exhibits the **law of diminishing marginal rate substitution**.

C) ICs don't intersect each other or tangent to one another. If they did, the point of their intersection would imply two different level of satisfaction, which is impossible. The intersection of ICs violates the consistency and transitivity rule which is logical necessity in IC analysis.

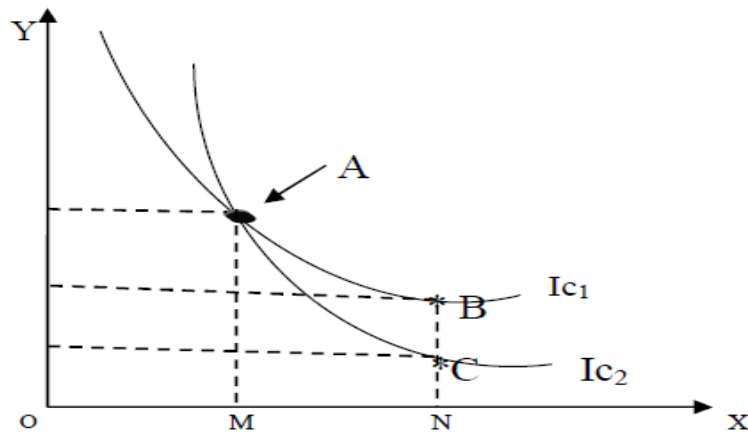


Fig 1.7

From the above fig 1.7 consumer satisfaction at point $A = B$, $A = C$. Therefore, based on the transitivity assumption $B = C$, but this is false because at point B consume $ONx + NB y$ and at point C consume $ONx + NC y$ so, $B \neq C$. The intersection of two ICs would mean that a commodity can be consumed either with a very much smaller or very much larger combinations.

- D) An indifference curve doesn't touch either axis. If it touch, it means that only one good is consumed and nothing of the other. This condition implies perfect substitutability.
- E) Higher IC represents a higher level of satisfaction than the lower one. In other words, more is preferred to less (*More is better* assumption of consumer preference). For example, in $IC_3 > IC_2 > IC_1$ as shown in the fig 1.8.

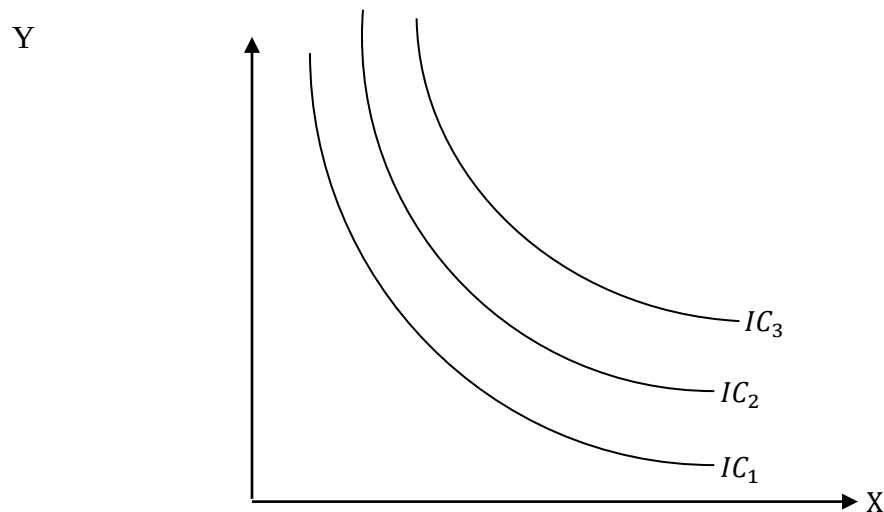


Fig 1.8

1.2.2.4. Special Cases of Indifference Curve

The shape of an indifference curve describes the willingness of a consumer to substitute one good for another. An indifference curve with a different shape implies a different willingness to substitute. There are two extreme cases of indifference curve. Such as,

Case 1: perfect Substitutes

If two goods X and Y are perfect substitutes, then the indifference curve will be a straight line (constant sloping downward) and intersecting the commodity axis. For goods X and Y to be perfect substitute, the MRS_{xy} must be constant. That means, no matter on what indifference curve we are, we must give up the same amount of good Y to get one additional unit of X .

Case 2: perfect Complements

If two goods X and Y are perfect complements to each other, they consumed together in fixed proportion. The indifference curve takes the shape of a **right angle (L-shaped)**. The MRS_{xy} will be zero in this case because there is no substitution possibility. These two extreme types of indifference curves are shown in panel (a) and panel (b) respectively. Panel (c) shows the normal indifference curve where we can have the possibility of substitution between two commodities.

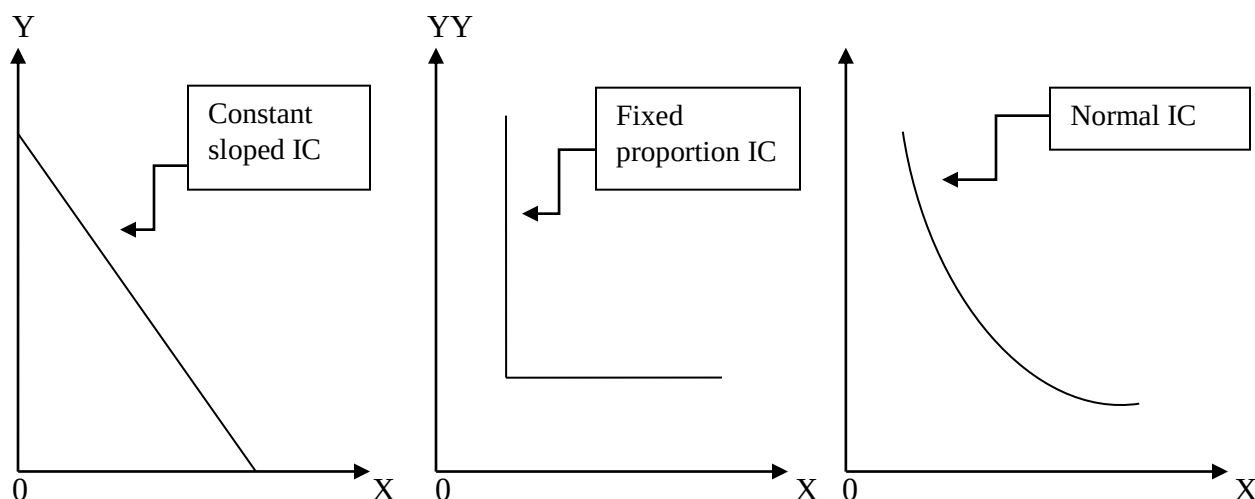


Fig 1.9: Types of Indifference Curves

Note: Greater the curvature of the indifference curves, lesser will the degree of substitution for goods. This reflects the law of $DMRS$. On the other hand, the curve is flatter (i.e. of less

curvature) the degree of substitution between goods will be very high and MRS will tend towards constancy.

Dear student! What influence the MRS will have on the shape of an indifference curve?

1.2.2.5. The Budget Constraint (Line)

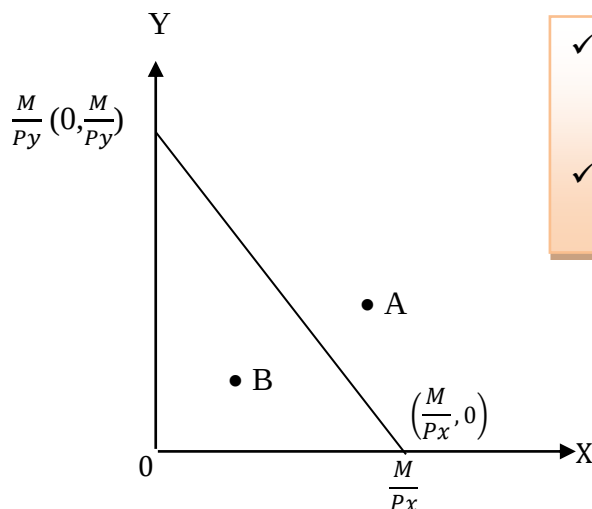
The limited income acts as a constraint to the quantities of goods the consumer can buy. This is known as *budget constraint*.

Budget constraint shows the various combinations of two commodities that a consumer can buy given his income and prices of the goods. The consumer's income limits the combinations of goods that the consumer can choose to consume. The combinations of the two goods that exhaust income are depicted by the budget line. The slope of the budget line represents the ratio of the prices of the two goods.

The income constraint in the case of two commodities can be written $P_x \cdot X + P_y \cdot Y \leq M$ where, the consumers purchase bundles (X and Y) at prices P_x and P_y respectively when the consumer has income M . The budget equation is $P_x \cdot X + P_y \cdot Y = M$. Then,

$$P_y \cdot Y = M - P_x \cdot X$$

$Y = \frac{M}{P_y} - \frac{P_x}{P_y} \cdot x$ By using this formula, we can draw the budget line.



- ✓ Point "A" is unattainable, at this point the consumer have limited income to purchase the goods.
- ✓ Point "B" is attainable; at this point the consumer doesn't spend all of his income.

Fig 1.10: Budget Line

$$\text{Slope of budget line} = \frac{\Delta y}{\Delta x} = \frac{0 - \frac{M}{P_y}}{\frac{M}{P_x} - 0} = -\frac{P_x}{P_y} \longrightarrow \text{The -ve sign show the negative (down ward) slope of the budget}$$

- ☞ The slope of the budget line ($-\frac{P_x}{P_y}$) is show the market willingness not the consumer willingness to substitute one good for another.
- ☞ The Y-intercept (Vertical intercept or $\frac{M}{P_y}$) shows all the consumer income spent on the consumption of good Y.
- ☞ The X-intercept (Horizontal intercept or $\frac{M}{P_x}$) shows all the consumer income spent on the consumption of good X.

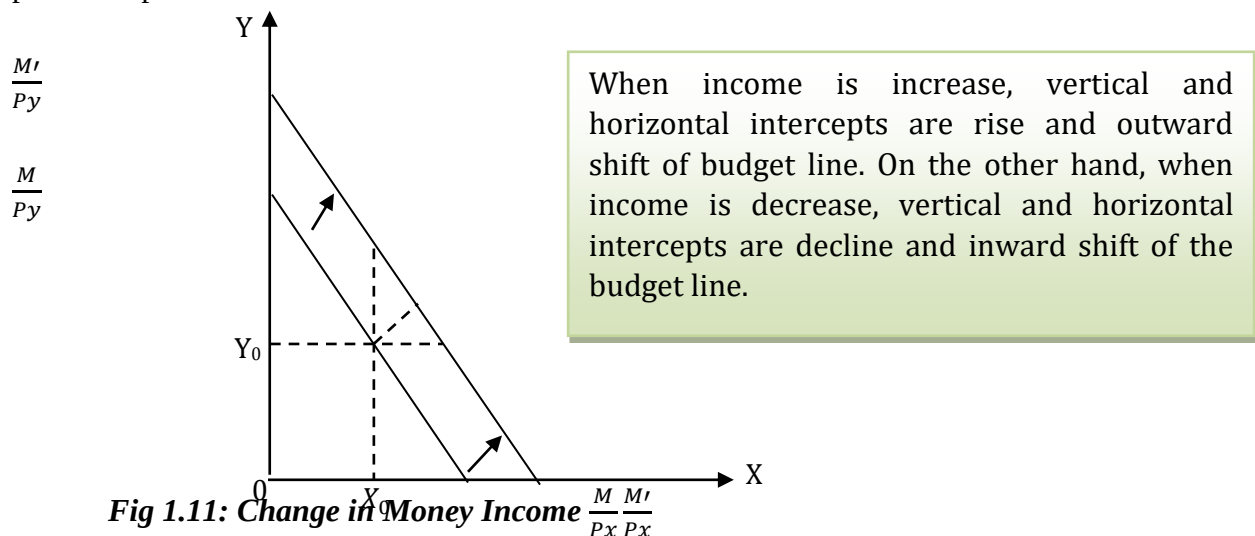
❖ Shift in the Budget Line

The budget line changes its position following the change in consumer's income and prices of commodity.

? *Dear student! How do the prices and the income affect the shape of the budget line? What happens if the price of one good rises? What happen if income increases?*

a) Change in Money Income

Increase in income of the consumer, P_x & P_y remaining the same, causes the budget line to shift upwards parallel to the original budget line. While the vertical and horizontal intercepts are increases. This type of change doesn't affect the slope of the budget line because income is not a part of slope. The inverse is also true.



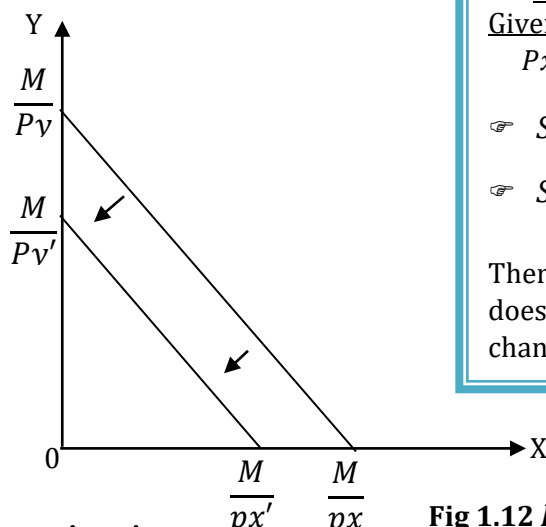
b) Change in Price

The slope and the intercepts of the budget line are affected by the changes in the price of the commodities. The change in price could have two types:

- i) Proportional change in price
- ii) Relative change in price

i) Proportional change in price

Proportional rise in the price of both goods, other things remaining unchanged, will reduce the total quantity of the two goods that the consumer can buy for the given income, and the vertical & horizontal intercepts are decline but leaves the slope of the budget line unchanged. Similar to price fall.



Calculating slope of the budget line

Given:

$P_x \uparrow = 2 \rightarrow 4$ and $P_y \uparrow = 3 \rightarrow 6$, then

$$\text{Slope}(\text{before } P \text{ rise}) = \frac{P_x}{P_y} = \frac{2}{3}$$

$$\text{Slope}(\text{after } P \text{ rise}) = \frac{P_x}{P_y} = \frac{4}{6} = \frac{2}{3}$$

Therefore, the slope of the budget line doesn't change due to the proportional change in price.

Fig 1.12 Proportional change in price

ii) Relative change in price

It can occur as the result of change in one of the prices or change in both prices in different portions/direction (and magnitudes). In this case both slope and intercept affected by the change.

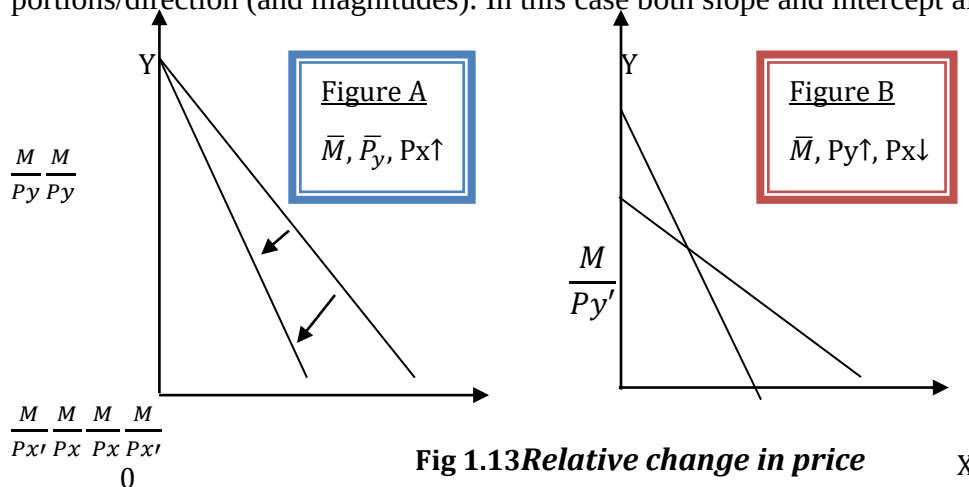


Fig 1.13 Relative change in price

1.2.2.6. Consumer Equilibrium (Under Ordinal Utility Approach)

Two conditions must be fulfilled for the consumer to be in equilibrium

- 1) The slope of the IC must be equal to the slope of the budget line

$$\frac{Px}{Py} = \frac{MUx}{MUy} (MRS_{xy}) \rightarrow \text{this is necessary condition}$$

- 2) The IC must be convex to the origin

Graphical presentation of the equilibrium

Given the indifference map of the consumer and his budget line, the equilibrium is defined by the point of tangency of the budget line with the highest possible indifference curve (Point “E”).

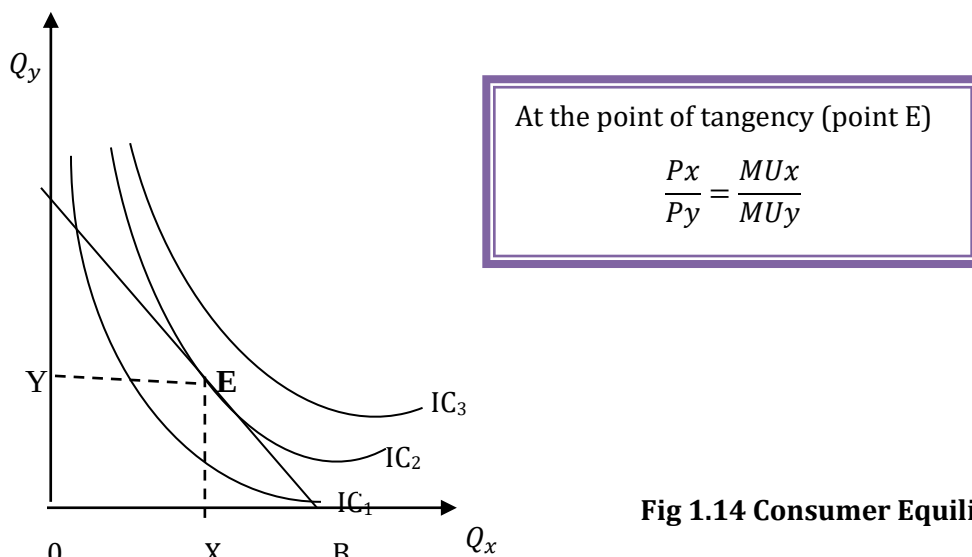


Fig 1.14 Consumer Equilibrium

Mathematical Derivation of the Equilibrium

Assume that there are “n” numbers of commodities with price $P_1, P_2, P_3, \dots, P_n$ and the consumers income is M . Formally the problem may be stated as

Maximize: $U = F(Q_1, Q_2, Q_3, \dots, Q_n) \rightarrow$ Objective function

Subject to: $P_1Q_1 + P_2Q_2 + \dots + P_nQ_n = M \rightarrow$ Constraint function

We use the lagrangian multiplier method for the solution of this constrained maximum

- 1) Rewrite the constraint: $P_1Q_1 + P_2Q_2 + \dots + P_nQ_n - M = 0$

- 2) Multiply the constraint function by lagrangian multiplier “ λ ”:

$$\lambda (P_1Q_1 + P_2Q_2 + \dots + P_nQ_n - M)$$

- 3) Forming the lagrangian function by subtracting the above constraint from the objective function: $L = U - \lambda (P_1Q_1 + P_2Q_2 + \dots + P_nQ_n - M)$

The Lagrangian theorem states that the optimum of the consumer should fulfill the following three first order conditions.

$$\frac{\partial L}{\partial Q_1} = \frac{\partial U}{\partial Q_1} - \lambda P_1 = 0 \text{ Condition 1.} \longrightarrow$$

$$\frac{\partial L}{\partial Q_2} = \frac{\partial U}{\partial Q_2} - \lambda P_2 = 0 \text{ Condition 2.} \longrightarrow$$

$$\frac{\partial L}{\partial \lambda} = -(P_1 Q_1 + P_2 Q_2 + \dots + P_n Q_n - M) = 0 \longrightarrow \text{Condition 3.}$$

$$= (P_1 Q_1 + P_2 Q_2 + \dots + P_n Q_n - M) = 0$$

$$= P_1 Q_1 + P_2 Q_2 - M = 0$$

$$\text{From (1)} \rightarrow MU_1 - \lambda P_1 = 0 \text{ then, } MU_1 = \lambda P_1 \rightarrow \frac{MU_1}{P_1} = \lambda$$

$$\text{From (2)} \rightarrow MU_2 - \lambda P_2 = 0 \text{ then, } MU_2 = \lambda P_2 \rightarrow \frac{MU_2}{P_2} = \lambda$$

$$\text{From (3)} \rightarrow P_1 Q_1 + P_2 Q_2 + \dots + P_n Q_n = M \text{ Then,}$$

$$\frac{MU_1}{P_1} = \lambda = \frac{MU_2}{P_2}, \text{ therefore, } \frac{MU_1}{P_1} = \frac{MU_2}{P_2}$$

We can rearrange in the form of: $\frac{MU_1}{MU_2} = \frac{P_1}{P_2}$

Slope of the budget line

Slope of indifference curve (MRS)

Example:

Suppose the utility function of a person consuming two commodities X and Y with income of 600 Birr is given by $U = 2X^{0.6}Y^{0.4}$. If the per unit price of X and Y is Birr 20 and 40 respectively. Then, find

- A) The utility maximizing level of consumption of X and Y?
- B) The MRS_{xy} at optimum?
- C) The maximum utility?
- D) The value of λ & interpret the value?

Solution:

Formally the problem may be stated as

$$\begin{aligned} \text{Max } U &= 2X^{0.6}Y^{0.4} \\ \text{Subject to } 20X + 40Y &= 600 \end{aligned}$$

A) In order to calculate the utility maximizing level of consumption of X and Y we should formulate the Lagrangian function

$$L = 2X^{0.6}Y^{0.4} + \lambda(600 - 20X - 40Y)$$

To precede the consumer should satisfy the following first order conditions

$$\frac{\partial L}{\partial X} = 1.2X^{-0.4}Y^{0.4} - 20\lambda = 0 \rightarrow \lambda = 0.06X^{-0.4}Y^{0.4} \dots \dots \dots (1)$$

$$\frac{\partial L}{\partial Y} = 0.8X^{0.6}Y^{-0.6} - 40\lambda = 0 \rightarrow \lambda = 0.02X^{0.6}Y^{-0.6} \dots \dots \dots (2)$$

$$\frac{\partial L}{\partial \lambda} = 600 - 20X - 40Y = 0 \dots \dots \dots (3)$$

From (1) and (2)

$$\begin{aligned} \lambda &= \lambda \\ 0.06X^{-0.4}Y^{0.4} &= 0.02X^{0.6}Y^{-0.6} \end{aligned}$$

After we rearrange we obtain

$$X = 3Y \dots \dots \dots (4)$$

Insert (4) in (3) we obtain

$$\begin{aligned} 600 - 60Y - 40Y &= 0 \\ Y &= 6 \text{ units} \end{aligned}$$

Insert the result of 'Y' commodity in (4) we obtain

$$X = 3Y = 18 \text{ units}$$

The above result show that the consumer can maximize his/her utility by consuming 6 units of commodity 'Y' and 18 units of commodity 'X'.

B) To compute the MRS_{XY} we should follow the following process

$$\text{We know that } MRS_{XY} = \frac{MU_X}{MU_Y}$$

Therefore, we should calculate MU_X and MU_Y to solve MRS_{XY} , so

$$MU_X = \frac{\partial U}{\partial X} = 1.2X^{-0.4}Y^{0.4} \quad \text{and} \quad MU_Y = \frac{\partial U}{\partial Y} = 0.8X^{0.6}Y^{-0.6}$$

At optimum,

$$MRS_{XY} = \frac{MU_X}{MU_Y} = \frac{1.2X^{-0.4}Y^{0.4}}{0.8X^{0.6}Y^{-0.6}} = \frac{1.2Y}{0.8X} = \frac{1.2(6)}{0.8(18)} = 0.5$$

The above result implies that the consumer should give up 0.5 units of commodity ‘Y’ in order to gain one unit of commodity ‘X’.

☞ **NB:** you should know the difference between MRS_{XY} and MRS_{YX} . And try to calculate the value of MRS_{YX} and compare the two results?

C) It is so simple to calculate the maximum utility by inserting the optimal amount of ‘X’ and ‘Y’;

$$\text{Max } U = 2X^{0.6}Y^{0.4} = 2(18)^{0.6}(6)^{0.4} = 23.198$$

The above result show that the consumer can drive the maximum satisfaction/utility 23.198 by consuming 6 units of commodity ‘Y’ and 18 units of commodity ‘X’.

D) The value of the Lagrangian multiplier, λ , can be manipulate by substituting the value of ‘X’ and ‘Y’ in to (1) or (2). Let’s take the first equation

$$\lambda = 0.06X^{-0.4}Y^{0.4} = 0.06(18)^{-0.4}(6)^{0.4} \approx 0.0387$$

Interpretation: It indicates that one unit change in the constant of the constraint will change the objective function by 0.0387 units.



Dear student! State the definitions of price elasticity, income elasticity, and cross-price elasticity of demand. What do these definitions mean in words?

1.3. Elasticity of Demand

Suppose we want to study the effects a price change has on the demand for a good. It is practical to do that in terms of percentages: if the price rises by one percent, how many percentages will demand change?

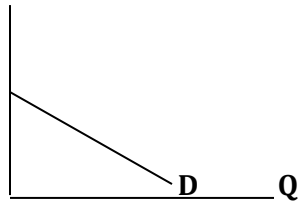
Elasticity of Demand is a measure of *responsiveness, or sensitivity*, of consumers to a change in one of the factors that affect demand (like price, income, price of other goods...). Accordingly, we have *price elasticity, income elasticity and cross-price elasticity of demand*.

(A) Price Elasticity of Demand: -it is a measure of responsiveness, or sensitivity, of consumers (i.e. their quantity demanded) to a change in the price of a product, other factors being the same.

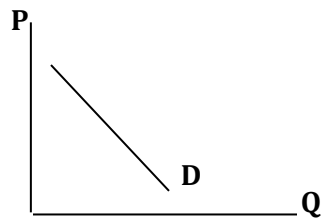
$$Ed = \frac{\% \text{ Change in Quantity Demand}}{\% \text{ Change in Price}}$$

$$\text{Symbolically, } Ed = \frac{\Delta Qd}{Qdo} \div \frac{\Delta P}{Po} = \frac{Qd - Qdo}{Qdo} \div \frac{P - Po}{Po} \implies Ed = \frac{Qd - Qdo}{P - Po} \cdot \frac{Po}{Qdo}$$

- a) Elastic Demand** -demand is said to be elastic if a specific percentage change in price results in a larger percentage change in quantity demanded. Then Ed will be greater than 1. Example: if a 3% decline in price results in a 9% increase in quantity demanded, then demand is elastic and $Ed = 0.09 / 0.03 = 3 > 1$. The demand curve in this case is drawn *flatter*. Most luxury goods & comforts have elastic demand.

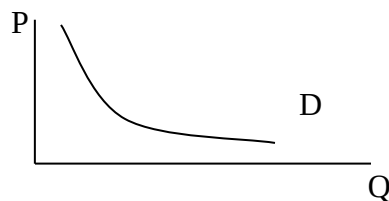


- b) Inelastic Demand:** -if a given percentage change in price is accompanied by a relatively smaller change in the quantity of the good or service, then demand is said to be inelastic. For example, if a 10% increase in the price of a product is accompanied by only a 2% decrease in the quantity demanded of that product $Ed = 0.02 / 0.1 = 0.2 < 1$. The demand curve in this case is drawn *steeper*. Most of the essential goods or necessities have inelastic demand. $0 < Ed < 1$

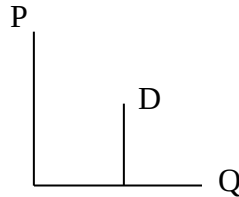


- c) Unitary Elastic:** - the case separating elastic and inelastic demands occurs where a percentage change in price and the accompanying percentage change in quantity demand are equal. For example if, a 6% change in price results in 6% change in quantity demanded, then $Ed = \frac{6\%}{6\%} = 1$.

In this case the demand curve is a *rectangular hyperbola*.



d) Perfectly Inelastic: - this is a situation the coefficient of elasticity, $E_d = 0$. This shows that a change in the price of a good or service does not bring in any change in, what so ever is, the quantity demanded (i.e. $(\Delta Q_d = 0)$). Some essential goods like medicine or goods having caffeine content pose perfectly inelastic demand. Typical examples are an acute diabetic patient's demand for insulin or an addicted person's demand for heroin. The demand curve for perfectly inelastic demand is a *vertical line* parallel to the price axis.



e) Perfectly Elastic: - this indicates that one percentage change in price results in infinite change in quantity demanded. In this regard, the consumer can buy all possible quantities at the given price and nothing else at some other prices. The demand curve for a perfectly elastic demand is a *horizontal line* drawn parallel to the quantity axis.

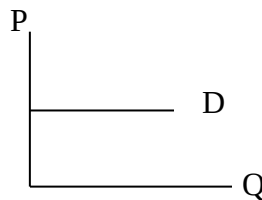


Table 1.2. Summary

Elasticity	Description	Implication	Demand curve
$E_d > 1$	Elastic	$\% \Delta Q > \% \Delta P$	Flatter
$E_d = 1$	Unitary elastic	$\% \Delta Q = \% \Delta P$	Rectangular Hyperbola
$0 < E_d < 1$	Inelastic	$\% \Delta Q < \% \Delta P$	Steeper
$E_d = 0$	Perfectly inelastic	$\% \Delta Q = 0$	Vertical
$E_d = \infty$	Perfectly elastic	$\% \Delta P = 0$	horizontal

From our definition of elasticity, $E_d = \frac{\% \text{change in quantity demanded}}{\% \text{change in price}} = \frac{\% \Delta \text{dependent variable}}{\% \Delta \text{independent variable}}$

However, we have to distinguish b/n **point and arc** elasticities. Point elasticity measures elasticity at a specific point on a demand curve, i.e., consumer responsiveness to price changes. Whereas, arc elasticity refers to price elasticity over a distance on the demand curve. In other words, arc elasticity measures the average responsiveness of consumer demand to changes in price over a range of extended prices.

$$\text{Point elasticity of demand} = \frac{\% \Delta \text{quantity demanded}}{\% \Delta \text{price}} = \frac{\Delta Q_d}{\Delta p} \cdot \frac{p}{Q_d}$$

$$\text{Arc elasticity of demand} = \frac{\% \Delta \text{quantity demanded}}{\% \Delta \text{price}} = \frac{\Delta Q_d}{\Delta P} \cdot \frac{P_1 + P_2}{Q_1 + Q_2}$$

The major **determinants** of price elasticity of demand are:

→ **Availability of close substitutes:** - a substitute for a product is one that serves the same general purpose. Therefore, the demand for a good that has many close substitutes is more elastic and the good that has few or no close substitute is inelastic.

→ **The proportion of income consumers spends on the commodity:** - the demand for a product on which a consumer spends large percentages of his/her income is likely to be quite elastic. Other things being equal, the smaller the percentage of income spent on a product, the less elastic the demand unless the good is considered as a disposable luxury.

→ **The nature of a commodity:** - the items that a consumer can use may be either necessary or luxury. In general, the demand for necessities is inelastic. This means that whatever be the changes in prices of such goods & services, certain quantities will still be purchased by consumers. However, if the item is a luxury good, the consumer may postpone his/her consumption, buy more when its price decreases or buy less when its price rises. Hence, demand is elastic for luxury goods. But, demand is inelastic in the case of addictive goods or alcoholic drinks. For example, if

an individual is addicted to cigarettes or alcoholic drinks, he/she finds it difficult to forgo though price increases and demand is highly inelastic.

→**Price level:** - Assuming that the demand curve for a product is linear, demand tends to be more elastic at higher price than at lower prices. This means moving along a linear demand curve, elasticity is not the same at every point. Even though a linear demand curve has constant slope, its elasticity varies from point to point.

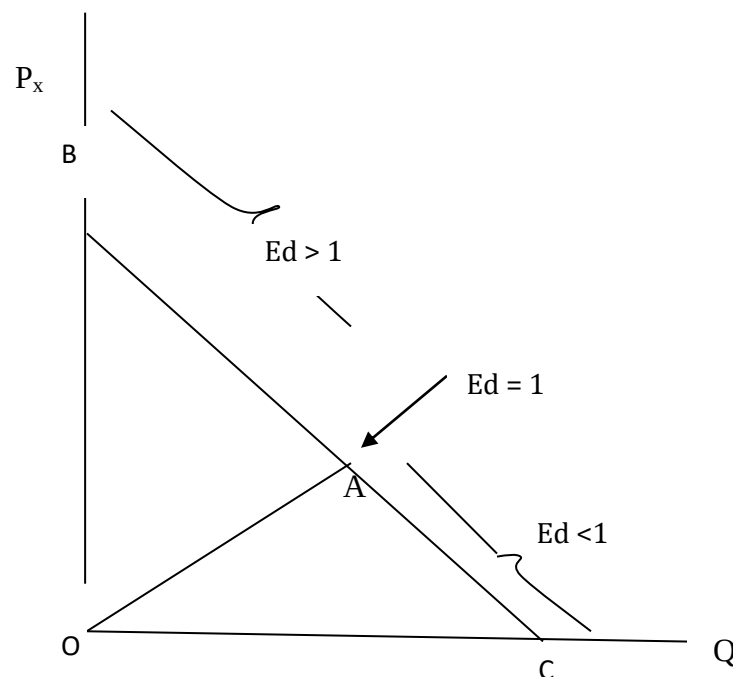


Fig 1.15 Elasticities on a linear demand curve

As shown in the above figure, along the, down-ward sloping demand curve, demand is elastic at higher prices, unitary elastic at the mid-point but becomes less elastic as price declines towards the lower end. At relatively low prices.

→**Adjustment time:** - In general, demand tends to be more elastic with longest time because the consumer can find more substitutes for the product in the long run. Over longer periods, we also have more time to adjust our consumption pattern in response to price changes which also contributes to a more elastic demand. Thus, in the short-run demand is inelastic while in the long-run it is elastic/relatively elastic.

Ex1. A 5% fall in the price of a good leads to a 20% rise in the quantity demanded.

Determine the elasticity and comment on its value.

Solution: $E_d = \frac{20\%}{5\%} = 4$ --- the demand is elastic

Ex2. With an 8% rise in the price of a commodity, the quantities demanded falls from 10 to 2 units. Determine the price elasticity of demand?

Solution $E_d = \frac{\% \Delta Q_d}{\% \Delta p} = \frac{2-10}{8\%} = \left| -\frac{80\%}{8\%} \right| = 10$ --- the demand is elastic

(B) Income Elasticity of Demand: - measures the responsiveness of demand of a particular commodity to changes in income of the consumer.

$$E_m = \frac{\% \Delta \text{quantity demanded}}{\% \Delta \text{income}}$$

This coefficient may be positive or negative. Goods that have positive income elasticity of demand are called **normal goods**. The quantity demand of such goods is positively associated with consumers' income. Whereas a negative income elasticity of demand implies an inverse relationship b/n income and the quantity of a good demanded. Such types of commodities are called **inferior goods**.

Normal goods may be either necessary or luxury items. We say a good is **necessity** good when $0 < E_m < 1$. But if $E_m > 1$, the commodity is **luxury**.

Table 1.3 Summary:

$E_m < 0$	$E_m > 0$	
The good is Inferior	The good is Normal	
	$0 < E_m < 1$	$E_m > 1$
	The good is Necessity	The good is luxury

C) Cross Elasticity of Demand: -Another useful price elasticity concept is the cross elasticity of demand, which measures the sensitivity of purchases of one goods say(X) to changes in the price of another good (say Y). We calculate the coefficient of cross elasticity of demand E_{xy} just as we do the coefficient of simple price elasticity, except that we relate the percentage change in the quantity demanded of commodity X to the percentage change in the price of Y:

$$E_{xy} = \frac{\% \Delta Q_x}{\% \Delta p_y} = \frac{\Delta Q_x}{\Delta p_y} \cdot \frac{p_y}{Q_x}$$

This coefficient tells us whether the two goods (X and Y) are substitutes, complements or independent (unrelated).

- i) **Substitute goods**: - If cross elasticity of demand is positive that is, the quantity demanded of X moves in the same direction as a change in the price of Y- then X and Y are substitute goods.
- ii) **Complementary goods**: - When cross elasticity is negative, we know that X and Y “go together” an increase in the price of one decrease the demand for the other.
- iii) **Independent goods**: - A zero or near-zero cross elasticity suggests that the two products are unrelated or independent goods.

Table 1.4

Summary:

$E_{xy} > 0$	$E_{xy} < 0$	$E_{xy} = 0$
Substitutes	complements	Independent

1.4. Income and Substitution Effects

Now we will now use consumer theory to perform a slightly more complicated analysis of a price change.

Substitution Effect: is the effect on the demand that depends on the change in relative price.

Income effect: is the effect on demand that depends on the fact that one can afford more after a drop in the price, and vice versa.

Suppose that we have a consumer, with a certain income, who has to choose between different quantities of good 1 and good 2 (which, again, can be thought of as “all other goods”) in such a way that she maximize her utility. If the price of good 1 falls, we get two different effects:

- ☞ Since the price of the good 1 falls, that good becomes cheaper relative the other good. This means that the marginal rate of substitution (MRS; slope of the budget line) changes. Say that the prices of both goods initially are 1. The relative price is then $1/1=1$. If the price of the good 1 falls to 0.50, the relative price becomes $0.50/1=0.50$. The consumer can now exchange one unit of good 2 for two units of good 1, and therefore good 1 becomes more

attractive to her. As a result, she consumes more of the good 1. This effect is called the **substitution effect**.

- ☞ The purchasing power of the consumer becomes larger because of the drop in the price. She can now buy as much as she did before the price changed, and still have money left. The extra money she can spend on both good 1 and on good 2. This is called the **income effect**.

1.5. Consumer Surplus

The consumer's surplus is a concept introduced by Marshall, who maintained that it can be measured in monetary units, and is equal to the difference between what consumers would have been willing to pay for each unit of a commodity and the amount they actually paid.

Graphically the consumer's surplus may be found by his demand curve for commodity x and the current market price. Assume that the consumer's demand for x is a straight line (AB in the Figure) and the market price is P. At this price the consumer buys q units of x and pays an amount (q) x (P) for it. However, he would be willing to pay p1 for q1, P2 for q2, P3 for q3 and so on. The fact that the price in the market is lower than the price he would be willing to pay for the initial units of x implies that his actual expenditure is less than he would be willing to spend to acquire the quantity q. This difference is the consumer's surplus, and is the area of the triangle PAC (the shaded area) in the figure.

To find the shaded area or the consumer surplus (Cs) in fig 1.16 we can use the triangle formula:

$$Cs = \frac{1}{2} (PA) \times (PC)$$

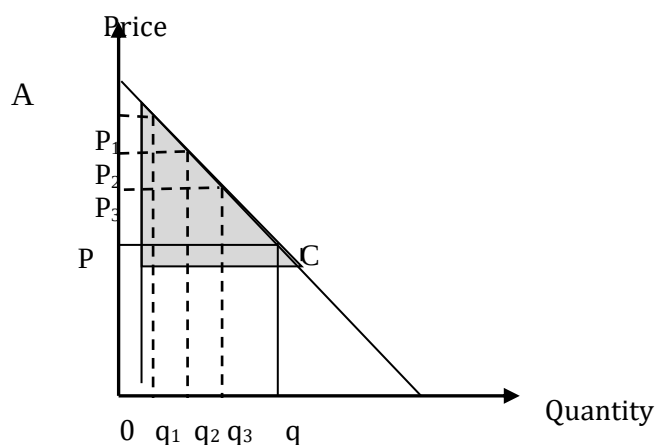


Fig 1.6. Consumer Surplus

Chapter- One: Review Questions

Part I: Choose the best answer from the given alternatives

- 1) Suppose the demand for banana sold from one roadside stand in Arba Minch is elastic. This fact means that a 7 percent increase in the price charged by the owner would lead to
- A) A greater than 7 percent increase in the quantity demanded at this stand.
 - B) A less than 7 percent increase in the quantity demanded at this stand.
 - C) A less than 7 percent decrease in the quantity demanded at this stand.
 - D) A greater than 7 percent decrease in the quantity demanded at this stand.
 - E) None of the above
- 2) Which one of the following is **not** the assumption of cardinal utility approach?
- A) Utility is quantitative feeling
 - B) Utility is measurable in objective units
 - C) Utility is subject to the law of diminishing marginal rate of substitution
 - D) Constant marginal utility of a unit of money
 - E) All except D
- 3) One of the following characteristics of indifference curve exhibited the law of diminishing marginal rate substitution concept
- A) Indifference curves are normally negatively sloped
 - B) Indifference curves are bowed inward to the origin
 - C) Indifference curves don't intersect each other or tangent to one another.
 - D) An indifference curve doesn't touch either axis.
 - E) None of the above
- 4) With an 8% rise in the consumer's income, the quantities demanded falls from 9 to 2 units. Which one of the following is true?
- A) The good is luxury
 - B) The good is necessity
 - C) The good is inferior
 - D) The good is Giffen
 - E) The good is normal
- 5) **Except one**, all of the following are **not necessarily true**?
- A) Transitive preference is not necessary for consumer consistency
 - B) If the good is undesirable, consumers always prefer more of any good to less
 - C) Due to the fall in price, if the consumer spend on/consume more of both good 1 and good 2

- D) Under cardinal utility analysis, when $MU_x > P_x$ the consumer can raise his/her satisfaction by consuming more units of good X
- E) None of the above
- 6) The Iso-utility curves of the consumer that lie far further from the quantity axis of the consumer represent
- A) Higher level of satisfaction C) The same level of satisfaction
- B) Lower level of satisfaction D) Higher level of profit E) B & D

Part II: Give the appropriate answer for the following questions

- 7) Suppose the budget line changes its position by the relative change in price of good “ X ” and good “ Y ”. If the price of the good “ X ” increase and the price of good “ Y ” decrease, assume money income is constant. Graphically show the effect of the price change on the position of the budget line?
- 8) Suppose the utility function of a person consuming two commodities X and Y with income of 480 Birr is given by $U = 50X^{0.5}Y^{0.5}$. If the per unit price of X and Y is Birr 20 and 40 respectively. Then, find
- A) The utility maximizing level of consumption of X and Y ? (NB: use the Lagrangian Multiplier Method)
- B) The MRS_{XY} and MRS_{YX} at optimum & interpret the result?
- C) The maximum utility?
- D) The value of λ & interpret the value?
- E) Calculate the maximum utility the consumer can obtain if his/her income increases by 25 Birr?
- 9) If the demand and supply functions of a good are given as $Q = 250 - 3P$ and $Q = 2P - 50$, respectively; then calculate the consumer surplus?
- 10) If the utility function of the consumer is given as: $U = \sqrt{XY}$, and the budget line equation is: $P_X \cdot X + P_Y \cdot Y = M$, then find the demand equation for X and Y ?

Chapter 2

2. Choice Under Risk

2.1. Introduction

So far, we have been implicitly assuming that information on prices and income is known by the consumer (decision maker). But many of the choice problems that the consumers face entail uncertainty about the possible outcomes of income price and other variables. Note that, information is a key input in decision making. Decision made under uncertainty of information involves some kind of risk.

Uncertainty is a situation which can result in a number of outcomes but one is not sure as to which outcome is to be materialized. On the other hand, risk is a consequence that one has to face as a result of the variability of outcomes from uncertain situation.

To analyze the choice behavior under uncertainty we need to quantify/ measure/ risk. To quantify risk, we need to know all the possible outcomes of an uncertain situation and the likelihood of occurrence of each of these events => Probability.

Ex- If we toss a fair die there are six equally likely outcomes. The probability that each side turn up is $1/6$.

Probability can either be objective or subjective. We say probability is objective, when probability is assigned based on observation with regard to the frequency at which the outcome occurred in the past. On the other hand, probability is called subjective probability, when probability is assigned based on personal assumption.

Utility Function under Uncertainty

If the consumer has reasonable preferences about consumption in different circumstances, a utility function can be used to describe these preferences. However, under conditions of uncertainty some additional structure, called probability, needs to be added to the choice problem.

Thus the utility function is defined based on the consumption levels as well as the probabilities of the state of nature. Let C_1 & C_2 represent consumptions in two states and let Π_1 & Π_2 be their respective probabilities, then we can write our utility function for the different consumption states as $U(C_1, C_2, \Pi_1 \text{ \& } \Pi_2)$.

2.2. Expected Utility

Expected Utility – is outcome expected on average.

Is the weighted average of all possible outcomes of an event (the weights being probabilities)?

Eg. Take an event that can result in two possible outcomes.

X_1 = Outcome 1

X_2 = Outcome 2

$E(X)$ = Expected Outcome

Let Π_1 is the probability of outcome 1 and

Π_2 is the probability of outcome 2, then $E(X) = \Pi_1 X_1 + \Pi_2 X_2$?

The formalization of consumer choice theory under uncertainty was initially by Neuman and Morganton. The central premise of the theory is that consumers choose the alternative with the highest expected utility rather than the highest value.

Like other utility functions, the expected utility function can be subject to monotonic transformation. The specific form of the utility function depends on the nature of individual consumer preferences. In consumption possibilities state 1 & 2 are perfect substitute $U(C_1, C_2, \Pi_1 \& \Pi_2) = \Pi_1 X_1 + \Pi_2 X_2$. The Cobb Douglas form is also possible. $U(C_1, C_2, \Pi_1 \& \Pi_2) = C_1^{\Pi_1} C_2^{\Pi_2}$. By taking the monotonic transformation we can re-write the utility function as $\Pi_1 \ln C_1 + \Pi_2 \ln C_2$. But the former representation doesn't have the expected utility property, while the latter does.

On the other hand, the expected utility function can be subject to some kinds of monotonic transformation and still have expected utility property. This special kind of monotonic transformation is called **positive affine transformation** which has the form of

$V(U) = au + b$, where $a > 0$. Positive affine transformation means multiplying the utility function by some positive number and adding a constant.

Economists argue that we can apply an affine transformation to expected utility and get another expected utility function that represents the same preferences. However, if any other kind of transformation is applied, it will destroy the expected utility property.

There are compelling reasons why expected utility is a reasonable for choice problems in the face of uncertainty. The fact that outcomes of the random choice are consumption goods that will be consumed in different circumstances means that ultimately only one of those outcomes is actually going to occur.

The consumption that will be available to the consumer under the two states of nature can be taken as independent with what the consumer will have if it didn't occur. And such additive form of expected utility function is acceptable for mutually exclusive uncertain outcomes.

If there are n outcomes with their respective probability, $C_1-\pi_1, C_2-\pi_2, C_3-\pi_3, \dots, C_n-\pi_n$, then the Expected utility will be $E(U) = \pi_1 C_1 + \pi_2 C_2 + \pi_3 C_3 + \dots + \pi_n C_n$

N.B. The expected utility function satisfies the condition that the marginal rate of substitution between the two goods is independent of how much there is of the 3rd good. For example $U(C_1, C_2, C_3)$ for a given utility function $MRS_{12} = MU_1/MU_2 =$

$$\frac{\frac{\Delta u(C_1, C_2, C_3)}{\Delta C_1}}{\frac{\Delta u(C_1, C_2, C_3)}{\Delta C_2}} = \frac{\pi_1 \Delta u(C_1)}{\Delta C_1} \bigg/ \frac{\pi_2 \Delta u(C_2)}{\Delta C_2}$$

Thus, from this we can conclude that $U(3)$ will not

be involved in the calculation of MRS_{12} .

Variability

Variability – refers to the event to which the individual outcomes are likely to vary from their expected value. We use variance or standard deviation to see the variation of individual outcomes from their expected value.

$$\text{Variance} = \sigma^2 = \pi_i \sum_{i=1}^n (X_i - \bar{x})^2$$

$$= \Pi_1 [X_1 - E(X_1)]^2 + \Pi_2 [X_2 - E(X_2)]^2 + \dots + \Pi_n [X_n - E(X_n)]^2$$

Choice with high variability results in high risk and similarly choice with low variability results in low risks.

Example

	Outcome 1		Outcome 2	
	Income	probability	Income	probability
Business 1	2000	0.5	1000	0.5
	1510	0.99	510	0.01

Expected income of business 1 $\Rightarrow E(1) = 0.5(2000) + 0.5(1000) = 1500$

Expected income of business 2 $\Rightarrow E(2) = 0.99(1510) + 0.01(510) = 1500$

$$\text{Variance} = \sigma^2 = \Pi_i \sum_{i=1}^n (X_i - \bar{x})^2$$

Variance 1 $= 0.5(250,000) + 0.5(250,000) = 250,000$

Variance 2 $= 0.99(100) + 0.01(980100) = 9900$

The outcome of the first business deviate from their expected value more than the second business and so is riskier. The preference of an individual will depend on his risk behavior.

2.3 Risk Aversion

Risk aversion is a tendency to prefer a given amount of income with certainty than taking a gamble with high possible return and possible loss. For risk averse person the utility from a given amount of income is greater than the expected utility from different levels of income with similar expected value.

Take a person whose utility depends on the level of income as follows. Suppose the person currently has 20,000 she is considering a business which will increase her wealth to 30,000 if the business succeeds but reduce her wealth to 10,000 if it fails.

Each success and failure have equal probability of 0.5. A utility for each state is given as follows.

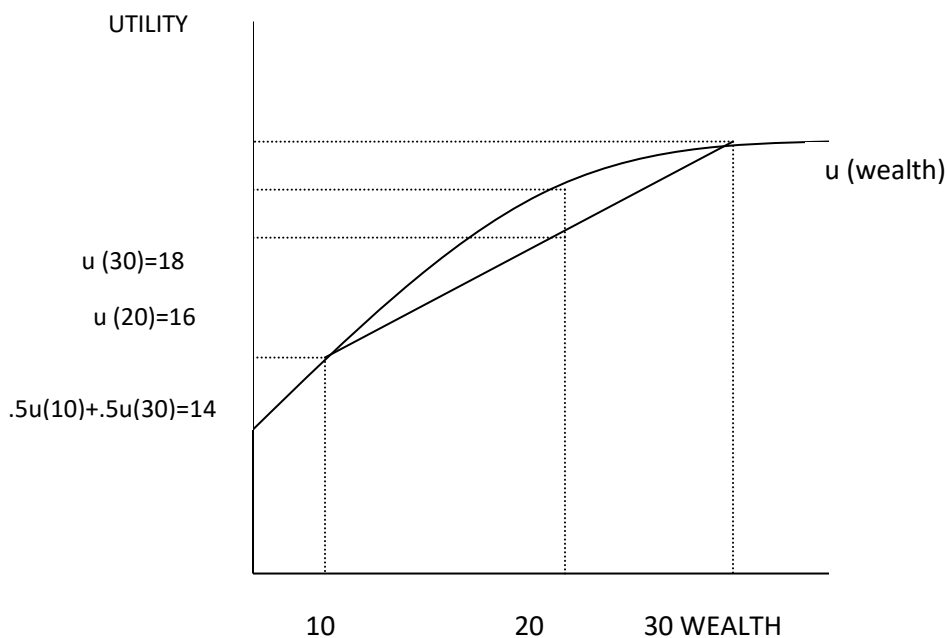
$$U(20,000) = 16, U(10,000) = 10, U(30,000) = 18$$

$$\begin{aligned}\text{Expected utility of wealth} &= 0.5U(30) + 0.5U(10) \\ &= 0.5(18) + 0.5(10) \\ &= 14\end{aligned}$$

$U(20) > E(U)$, for $U(20)$ is 16 and $E(U)$ is 14. Because of this the risk averse consumer accepts this gamble.

Note that: Risk aversion prefers a certain income to a risky income with the same expected value.

This is depicted in the following figure.



Risk Aversion. For a risk-averse consumer the utility of the expected value of wealth, $u(20)$, is greater than the expected utility of wealth, $.5u(10)+.5u(30)$.

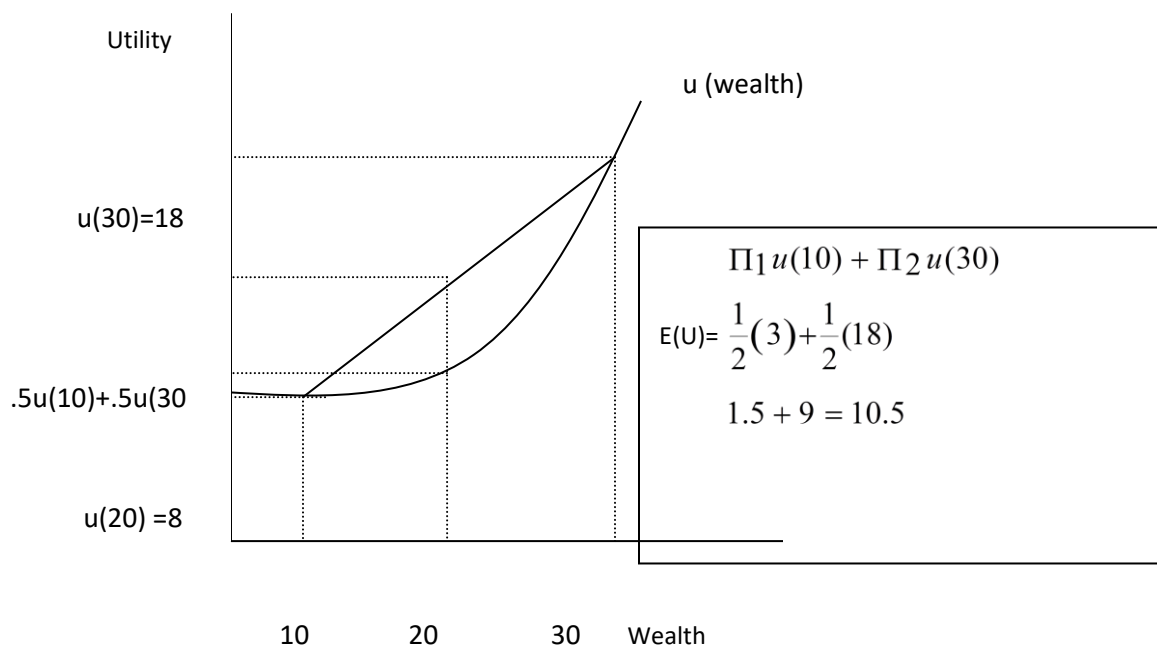
The risk averse consumer has a concave utility curve. This implies that the slope of a utility curve

gets flatter as wealth increases. The reason is that risk averse consumer has a diminishing marginal utility of income.

The marginal utility of income decreases with increasing level of wealth. This implies that the gain in utility from a given amount of additional wealth is smaller than the loss in utility from a comparable loss in wealth. That is why such a person always refuses to accept a gamble whose expected value is less than expected utility.

Risk Loving

This is a situation where the utility from a given amount of wealth (income) is less than the expected utility.

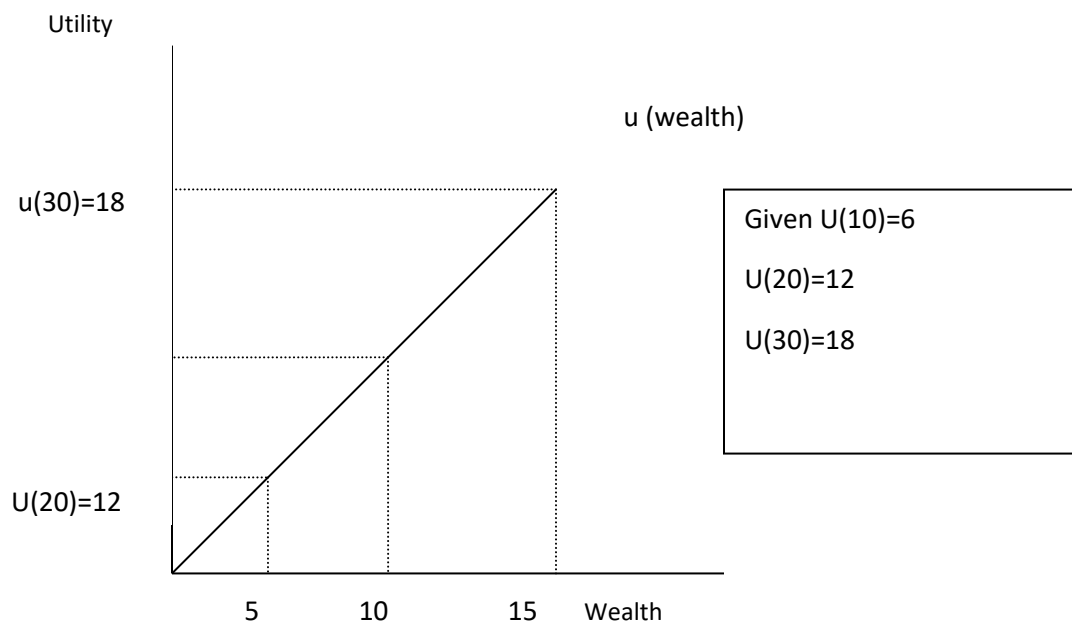


The risk lover consumer has a convex utility function. The slope of the utility function gets steeper and steeper as wealth increases. For such consumer, marginal utility increases with increasing level of wealth. This implies that the gain in utility from a given amount of additional wealth is greater than the loss in utility from a comparable loss in wealth. That is when such a person always accepts a fair gamble.

The curvature of the utility function measures the consumer's attitude towards risk. The more concave the utility curve, the more risk averse the consumer is. In contrast, the more convex the utility curve, the more risk lover the consumer is.

RISK NEUTRAL

If a person is indifferent between a given amount of wealth with certainty and expected value the person is said to be risk neutral.



A risk neutral consumer has straight line utility curve.

Exercise:

There are two outcomes if the person succeeds, he will get additional birr of 20, and if he loses he will lose 20 birrs. Each, loss and success, have 0.5 probability. Will the person accept the gamble, if he is risk neutral?

2.4 DIVERSIFICATION

Given that a consumer is a risk averse. He minimizes the risks of uncertainty, by diversifying his holding or assets. When one puts his effort in different types of (unrelated) activities, the loss from badly performing activities can be compensated by gain from well performing activities. By this method one can minimize his/her risk, this

method of minimizing risk is said to be diversification.

Ex: Consider a person thinking of investing Br. 1000 in two companies, one is heater producing and the other is air conditioners producing company. The share of stock for both companies currently is sold for Br. 100 each. The next season is equally likely to be hot or cold. If the next season turns out to be cold, the securities of the heaters company will be worth Br.200 each and securities of Air Conditioners Company will be worth of Br.50 each. And if the next season turns out to be hot the reverse will happen. Let's compare two cases:

1) If the person puts the whole investment in one of the companies say heater.

	Outcomes	Probabilities
Cold	200,000 (200X1000)	.5
Hot	500,00 (50X1000)	.5

2) If the person equally divides his investment and put in the two companies:
and if the weather turns out to be cold.

	Outcomes		
From heater	100,000	(200x500)	125,000 certain income
Air conditioner	25,000	(50x 500)	

If the weather turns out to be hot

From Air conditioner	100,000	(200x500)	125,000 certain income.
Heater	25,000	(50x500)	

2.5 RISK SPREADING

In the absence of formal insurance, a group of individuals can agree to share a loss incurred by each individual from their group.

Each consumer (individual) in a group spreads his/her risk over all of the other consumers and there by reduces the amounts of risk.

Ex:

Consider a situation of an individual who had Br.3500 and faced a 0.1 probability of Br.10,000 loss. Suppose that there were 1000 such individuals. On an average, 10 losses are expected to occur, thus Br.100,000 will be lost each year.

Review exercise on chapter two

1. The utility function of a consumer is given as $U=2W^{1/2}$ with an initial endowment of 60,000Birr. If he is involved in a game of fair gamble which could lead to a win/loss of 40,000Birr, determine the attitude of this consumer towards risk.
2. The utility function of a consumer is given as $U=2W^4$ with an initial endowment of 50Birr. If he has an equal chance of winning or losing 5Birr, determine the attitude of this consumer towards risk.
3. The utility function of a consumer is given as $U=W^{1/2}$ with an initial endowment of 900Birr. If he is involved in a game of fair gamble which could lead to a win/loss of 500Birr, determine how much the consumer is willing to pay to avoid risk.
4. Utility function is given by $1/3W$, initial endowment 600, and he/she has a 90% chance of winning 200 and 10% chance of winning 50
 - a. Is the consumer risk averse, risk lover or risk neutral
 - b. Find $E(W)$
 - c. Find $UE(w)$
 - d. Find $EU(W)$

CHAPTER THREE

THEORY OF PRODUCTION

Dear student! After successful completion of this unit, you will be able to

- Under the basic concept of production
- Define production and production function
- Compute the marginal and average product from the total production function
- Differentiate short run and long run, and fixed and variable inputs
- Know the concepts of short run production.
- Understand the concept of efficiency in production
- Understand the role of technology in production
- Know the concepts of long run production, laws of returns to scale and how to determine least cost production process.
- Determine the optimal input mix of a given production

3.1. Basic Concepts of Production

3.1.1. Production and Production Function

Production: is the process of using economic resources (factors of production) to make the final goods and services with value added. In other words, it is the transformation of inputs into outputs. The outputs can be tangible or intangible.

Production Function: is the technical and technological relationship between inputs and outputs. In other terms, a production function shows the maximum output that can be obtained from a given set of inputs. It describes not only a single isoquant but also the whole array of isoquants.

Symbolically, we can write this as follows:

$$Q = f (L_B, L, K, M, T, E, etc)$$

3.1.2. Classification of Inputs

The classification of economic resources as ***fixed*** and ***variable inputs*** is related with the concept of short run and long run. This classification is mainly made depending on the degree of flexibility of economic resources to changes in business environment of products consuming those resources.

Fixed inputs: one that is used in fixed quantities for a certain level of output, i.e. its quantity does not change with the change in output. The supply of such resources is inelastic in the short run and therefore, a production system cannot employ more of it in the short run. Example: Machinery, buildings, etc.

Variable inputs: are inputs whose quantity can be readjusted in relation to the level of production; or these inputs are resources whose quantity changes with the level of output, e.g., labor and raw materials. In the short run the supply of these inputs is elastic and therefore, it is possible to employ more of them in the short run production process.

3.1.3. The Short- Run and Long-Run Production Period

In economics, production period is classified as short run and long run.

Short- run production period: refers to a period of time in which the quantity of at least one of the inputs or resources is fixed. For example, capital goods like building, machinery, etc are fixed in the short run. In the short-run, production constitutes both variable and fixed inputs.

Long- run production period: on the other hand, refers to a period of time in which all inputs are variable or there is no fixed resource in general.

It must be understood that when we mean short run and long run it does not necessarily mean a relatively short or long period of time like one year or less than one year or like two or five years. It rather refers to the nature of economic arrangement of the inputs in response to the changing economic environment.

3.2. Production With One Variable Input

This is a phenomenon of short-run production period. The short-run decision of production is a time period over which production is carried out with the help of two categories of inputs i.e. fixed inputs and variable inputs. For simplicity purposes, we assume production is carried out

with capital (K) as a fixed input and labor (L) as a variable factor. Therefore, the production function can be written as:

$$Q = f(\bar{K}, L) \text{ or } Q = f(L)$$

3.2.1. Total, Average and Marginal Product

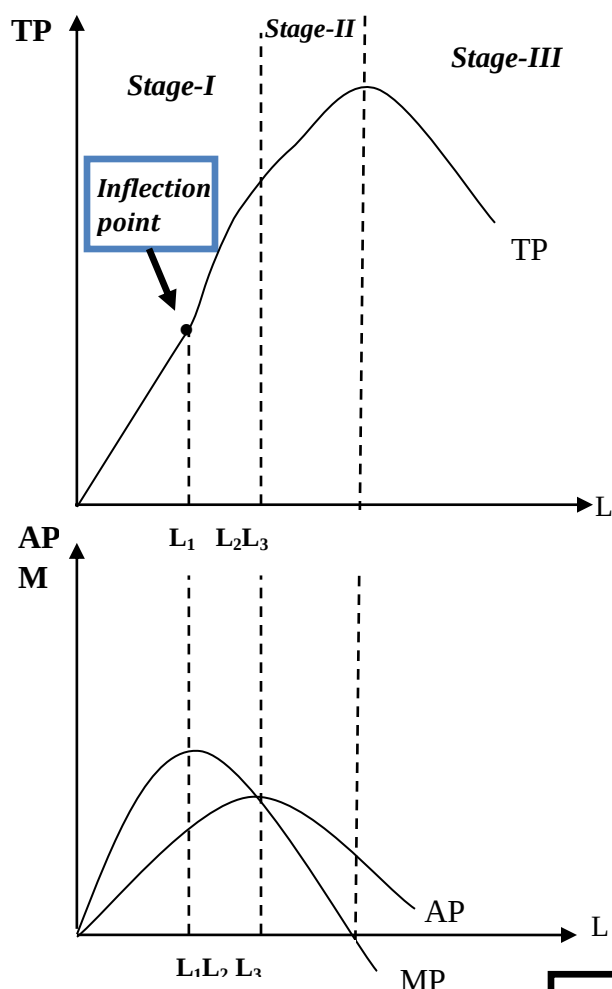
- A) **Total Product (TP)**: it is the overall amount of output produced by the factors of production employed over a given period. It is the gross or entire output by workers.
- B) **Average Product (AP)**: it is the ratio of total product to the amount of variable factor. This can be put in the form of; $AP_L = \frac{TP}{L}$. The average product is a good indicator of the *productivity* of labor. Productivity is a measure of output per unit input.
- C) **Marginal Product (MP)**: it is the increase in output which results from using one additional or extra unit of a single factor input, holding the quantities of other factors constant, is called the marginal physical product or simply marginal product. In other words, the MP is the percentage change in total output resulting from a percentage change in variable input or it is the additional output results from using one more unit of factor of production, all other things being equal. Mathematically, $MP_L = \frac{\partial TP}{\partial L} = \frac{\Delta TP}{\Delta L}$. It is the slope of total product.

3.2.2. The Law of Diminishing Marginal Returns (LDMR)

It is sometimes known as the “*law of variable Proportion*” or the “*law of returns to variable inputs*”. The law states that if some factors are held constant and more units of variable factors are employed, the marginal product of the variable input eventually decreases. In other words, it states that increasing the amount of the variable factor (labor) with fixed factor (capital) will lead to an eventual decline in the marginal contribution of the additional labor to the total output. This is also accompanied by the eventual decline in total output.

3.2.3. The Stages of Production

A functional relationship between observed between total product (TP), average product (AP) and Marginal product (MP) help to describe stages in production function (see fig. 2.1) states that as more of variable input is applied, output(Q) increases until the maximum TP at which input usage L_3 is reached. Further application of variable input beyond L_3 will only serve to reduce the total output.



Some Important points in the figure

- ☞ When MP reaches its maximum, TP reaches its **inflection point**.
- ☞ When MP increases, TP increases at increasing rate until its inflection point. Slope of TP is positive.
- ☞ When MP declines until MP=0, TP increases at decreasing rate.
- ☞ When MP=0, TP reaches its maximum (constant level).
- ☞ When TP start to decline, MP negative. Slope of TP is negative.
- ☞ The point at which MP attains its maximum is known as the point of LDMR. In other words, the LDMR begins to operate when MP begins to fall.
- ☞ The Point at which MP=AP is the point of diminishing average return (DAR).

Fig 2.1 Stages of Production

Stage – I: it is a range of increasing average physical product. The stage goes from the origin where $TP=0$ to the point where AP_L is maximum ($AP_L=MP_L$). It indicates the range of rising average physical product for the variable input (labor).

- ✓ It is largely stage of increasing returns to L, because TP increase at increasing rate.
- ✓ Variable inputs remain under employed & fixed inputs remains underutilized.
- ✓ $MP_L > AP_L$.

Stage – II: it is the stage of production ranges from maximum AP_L to zero MP_L . In other words, the stage two production begins when AP_L starts to decline and ends when MP_L curves becomes zero, or TP reaches its maximum level.

- ✓ It is the stage of diminishing returns because TP increase at decreasing rate.
- ✓ $MP_L < AP_L$.

- ✓ Both MP_L & AP_L are diminishing but +ve. MP_L decline at faster rate than AP_L .
- ✓ It is optimum stage of production.
- ✓ There is proper utilization and employment of resources.

Stage – III: it is the stage of production where the marginal contribution of any additional labor becomes negative or it is the level of production where the TP starts to decline with one more labor employed. This stage implies that labor is over employed and capital is over utilized. This stage of production is known as **intensive margin**. This ranges from $MP_L=0$ to through the –ve part of MP_L .

- ✓ It is the stage of negative returns because TP declines and MP_L decline.
- ✓ AP_L continues to decline though it remains positive.
- ✓ Both MP_L & AP_L are diminishing but $AP_L > 0$ and $MP_L < 0$.

A rational producer would not produce in stage-I because there is excess capital for the very limited variable input (labor). In stage I labor is underemployed and capital is underutilized. This stage of production is known as **extensive margin**. And also, rational producer would not produce in stage-III because there is overutilization of capital and labor is over employed. Therefore, the only stage production can take place is the stage II where both MP_L and AP_L are declining but are showing positive values. In other words, the only feasible stage of production for a rational producer is stage II.

Relationship between MP_L and AP_L

- ✓ The ascending and descending pattern of both AP_L and MP_L curves is similar.
- ✓ When $MP_L > AP_L$, AP_L is increasing
- ✓ When $MP_L < AP_L$, AP_L is decreasing
- ✓ When $MP_L = AP_L$, AP_L reaches maximum

Relationship between MP_L and TP_L

- ✓ When MP_L is increasing, TP is also increasing at increasing rate.
- ✓ When MP_L is decreasing until it reaches zero, TP is increasing but at decreasing rate.
- ✓ When $MP_L = 0$, TP reaches its maximum (saturation point or constant level).
- ✓ When MP_L is negative, TP is declining.

Example for Short- run production

Table 2.1 short run productions

Fixed input	Variable	Total	Average	Marginal	Stages of
-------------	----------	-------	---------	----------	-----------

(a hectare of land)	input (labor)	Product (TP) (In quintals)	product (AP)	Product (MP)	Production
1 hectare	0	0	-	-	Stage I
»	1	10	10	10	
»	2	28	14	18	
»	3	51	17	23	
»	4	76	19	25	
»	5	95	19	19	Stage II
»	6	108	18	13	
»	7	108	15.4	0	
»	8	96	12	-12	Stage III
»	9	80	8.88	-16	

As it can be seen from the table, the total product goes through three different stages. I hope that the above graphical representation of the production function makes your understanding deeper as the graph has three different natures of steepness showing the corresponding stages of production. It also clearly explains the law of diminishing marginal returns.

3.3. Production With Two Variable Inputs

This is a phenomenon of long-run production period. It is a production period over which all factors of production are variable. For simplicity purposes, we assume production is carried out with both capital (K) and labor (L) as a variable factor. Therefore, the production function can be written as:

$$Q = f(K, L)$$

3.3.1. Isoquant Analysis of Production

The long-run production period can be analyzed using Isoquants. An ***Isoquant (Isoproduct)*** is a curve or a line which the various combinations of variable inputs yield the same level of output. In our case the combination of capital (K) and labor (L). In other words, an isoquant is a curve along which the maximum achievable rate of production is constant.

Characteristics of Isoquants

- A) Isoquants are normally negatively sloped: because when one factor increases, the other factor decreases.
- B) An isoquant laying above and to the right of another represents a higher level of output.

K

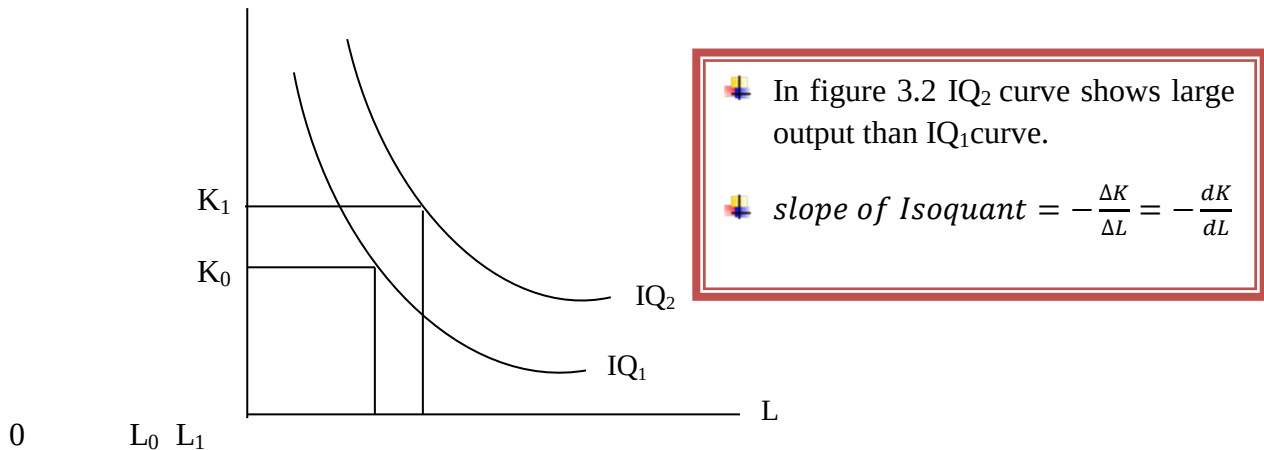


Fig 2.2 Two Isoquant

- C) Two isoquants cannot intersect each other or tangent to one another. If it is done, the various combination of variable inputs can produce the same level of output at the same time (at one point). Which is impossible.
- D) No isoquants can touch either axis: if an isoquant touches the x-axis, it would mean that the product is being produced with the help of labor alone without using any capital at all and vice versa.
- E) Each isoquant is normally drawn convex to the origin: as more units of labor are employed to produce, lesser and lesser units of capital used. This is so, because the marginal rate of technical substitution (MRTS) between the two factors diminishes.

3.3.2. The Marginal Rate of Technical Substitution (MRTS)

MRTS is the rate at which one input can be substituted for another, keeping the level of output constant.

$MRTS_{L, K}$ is the number of K which can be replaced/ substituted by one unit of labor, while $\overline{TP}$.

We are reducing the amount of K by fewer and fewer for each extra unit of labor. This is happens due to two reasons:

1. No factor of production is perfect substitute for another.
2. Inputs are subject to diminishing marginal return.

Mathematically,

$$MRTS_{L,K} = -\frac{\Delta K}{\Delta L} = -\frac{dK}{dL} = \frac{MP_L}{MP_K}$$

$-\Delta K MP_K = \Delta L MP_L$ The amount of loss is must equal to the amount of gain.

Proof

Given $Q = f(L, K)$

$$dQ = \frac{\partial Q}{\partial L} \cdot dL + \frac{\partial Q}{\partial K} \cdot dK$$

$$0 = MP_L \cdot dL + MP_K \cdot dK$$

$$MP_L \cdot dL = -MP_K \cdot dK$$

$$-\frac{dK}{dL} = \frac{MP_L}{MP_K} = MRTS_{L,K} \rightarrow \text{therefore, MRTS is the slope of Isoquant} \left(\frac{dK}{dL} \right)$$

In the same manner:

$$-\frac{dL}{dK} = \frac{MP_K}{MP_L} = MRTS_{K,L}$$

3.3.3. Elasticity of Substitution

Elasticity of substitution defined as the ratio of percentage change in the capital- labor ratio to the percentage change in the MRTS. In other words, it measures the degree of factors substitutability.

$$\delta = \frac{\text{percentage change in } \frac{K}{L}}{\text{percentage change in MRTS}} \rightarrow \delta = \frac{\% \Delta \left(\frac{K}{L} \right)}{\% \Delta (MRTS)}$$

$$\delta = \frac{d \left(\frac{K}{L} \right) / \left(\frac{K}{L} \right)}{d(MRTS) / MRTS} \quad \text{where, } \% \Delta \left(\frac{K}{L} \right) = \frac{\Delta \left(\frac{K}{L} \right)}{\left(\frac{K}{L} \right)} = \frac{d \left(\frac{K}{L} \right)}{\left(\frac{K}{L} \right)} \quad \text{and}$$

$$\% \Delta (MRTS) = \frac{\Delta (MRTS)}{MRTS} = \frac{d(MRTS)}{MRTS}$$

$$\delta = \frac{d\left(\frac{K}{L}\right)}{d(MRTS)} \times \frac{MRTS}{\left(\frac{K}{L}\right)}$$

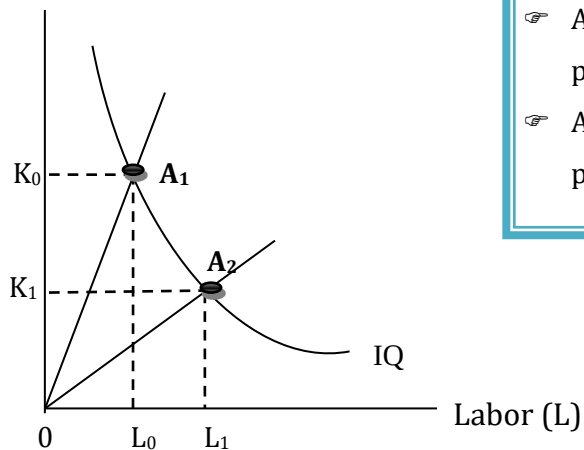
$\delta = 0 \rightarrow$ For perfect complement inputs

$\delta = \infty \rightarrow$ For perfect substitute inputs

$\delta = 1 \rightarrow$ For imperfect substitute inputs

The factor intensity of any process is measured by the ratio of capital and labour $\left(\frac{K}{L}\right)$

Capital (K)



☞ At point A₁, $K > L$ then the production process is **Capital intensive**.

☞ At point A₂, $L > K$ then the production process is **Labor intensive**.

Fig 2.3 Factor Intensity

3.3.4. Returns to Scale (law of production in the long run)

The term *returns to scale* refers to the changes in output as all factors change by the same proportion.

Suppose we start from an initial level of inputs and output.

$$Q = f(L, K)$$

And we increase all the factors by the same proportion 't', we will clearly obtain a new level of output $h(Q)$, higher than the original level Q ,

$$h(Q) = f(tL, tK)$$

☞ If $h(Q)$ increases by the same proportion 't' as the inputs ($h=t$), we say that there is **constant returns to scale (CRS)**.

☞ If $h(Q)$ increases less than proportionally with the increase in the factors ($h < t$), we have **decreasing returns to scale (DRS)**.

☞ If $h(Q)$ increases more than proportionally with the increase in the factors ($h > 1$), we have **increasing returns to scale (IRS)**.

1. **Increasing Returns to Scale (IRS)**: occurs when a percentage change in all inputs caused a more than percentage change in output. ($\% \Delta input < \% \Delta output$)

Reasons for IRS: - technical and managerial indivisibility

- Increase in the scale of operation
- Higher degree of specialization

2. **Constant Returns to Scale (CRS)**: occurs when a percentage change in all inputs leads to the same proportionate change in output. ($\% \Delta input = \% \Delta output$)

3. **Decreasing Returns to Scale (DRS)**: occurs when a percentage change in all inputs leads a less than proportionate change in output. ($\% \Delta input > \% \Delta output$)

☞ Returns to scale and Homogeneity of the Production Function

When we multiply all factors of production by the same proportion “ t ”, “ t ” can be factored out from the function, it is homogeneous production function.

Given: $Q = f(L, K)$

Suppose we increase both factors of production by the same proportion “ t ”, and we observe the resulting new level of output:

$$Q^* = f(tL, tK)$$

If “ t ” can be factored out (that is may be taken out of the brackets as a common factor), then the new level of output Q^* can be expressed as a function of “ t ” and the initial level of output

$$Q^* = t^v f(L, K) \text{ or } Q^* = t^v Q$$

And the production function is called homogeneous production function. If “ t ” cannot be factored out, the production is non-homogeneous. Thus, a homogeneous function is a function such that if each of the inputs is multiplied by “ t ”, then “ t ” can be completely factored out of the function. The power “ v ” of “ t ” is called the degree of homogeneity of the function and is a measure of the returns to scale.

☞ If $v = 1$, we have constant returns to scale. This production function sometimes called linear homogeneous production function.

☞ If $v > 1$, we have increasing returns to scale.

Given: $Q = 5L^{0.5}K^{0.2}$, State the returns to scale of the production function?

Solution: $Q^* = 5(t_L)^{0.5}(t_K)^{0.2}$

$$Q^* = 5t^{0.5}L^{0.5}t^{0.2}K^{0.2}$$

3.3.5. Isocost Curve/ Isoclines

i.e. $C = wL + rK$ Where, w- price of labor (wage) and r – price of capital (rent)
 L- Labor input and K- Capital input

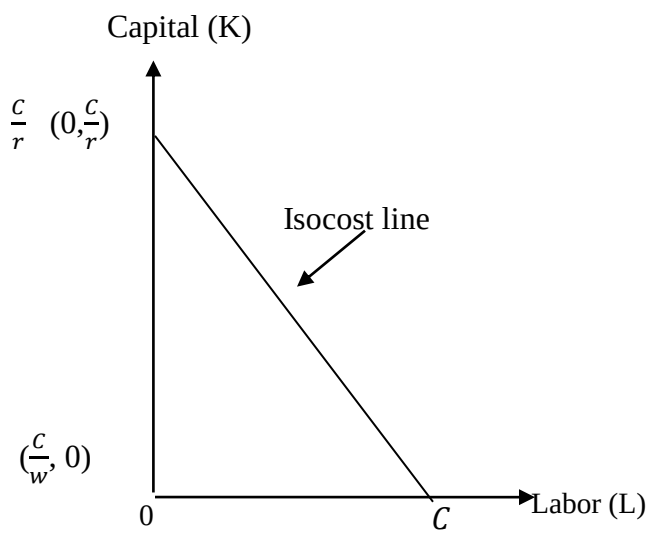
$$K = \frac{c}{r} - \frac{w}{r}L$$
 By using this formula, we can draw the Isocost line.

Fig 2.4 An Isocost

$$\text{Slope of Isocost line} = \frac{\Delta y}{\Delta x} = \frac{0 - \frac{C}{r}}{\frac{C}{w} - 0} = -\frac{w}{r}$$

The -ve sign show the negative (downward) slope of the Isocost line.

- ☞ The Y-intercept (Vertical intercept or $\frac{C}{r}$) shows the maximum amount of capital can purchase by the given cost. In other words, the firm's expenditure is only on capital.
- ☞ The X-intercept (Horizontal intercept or $\frac{C}{w}$) shows the maximum amount of labor can hire by the given cost. In other words, the firm's expenditure is only on labor.

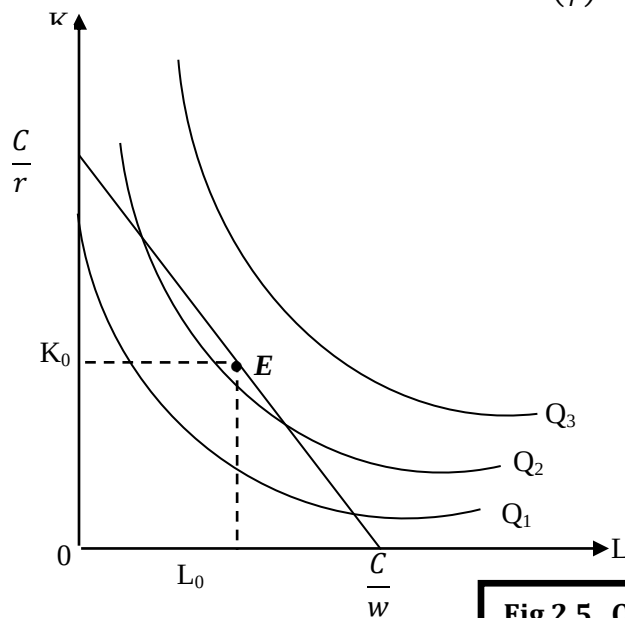
3.3.6. Choice of Optimal Combination of Inputs (Producer Equilibrium)

The optimal input combination of a firm depends on the ultimate goal of the producer. It is known that the main goal of the producer is to maximize output given a specified expenditure or to minimize the cost of production for producing a specified level of output.

Based on the above fact, the producer equilibrium can be described as:

A) Output Maximization for a given level of cost

The producer will be in equilibrium when he/she maximizes output for a given cost. This means that the producer reaches the highest isoquant for a particular cost. This occurs where the isoquant curve is just tangent to the isocost line (at point "E"). At this point, the slope of the isoquant (MRTS) = slope of isocost line $\left(\frac{w}{r}\right)$. i.e. $\frac{MP_L}{MP_K} = \frac{w}{r}$



Mathematically

Maximize $Q = f(L, K) \rightarrow$ Objective function

Subject to $C = wL + rK \rightarrow$ Constraint function

Then, the lagrangian function is:

$$L = f(L, K) - \lambda(wL + rK - C)$$

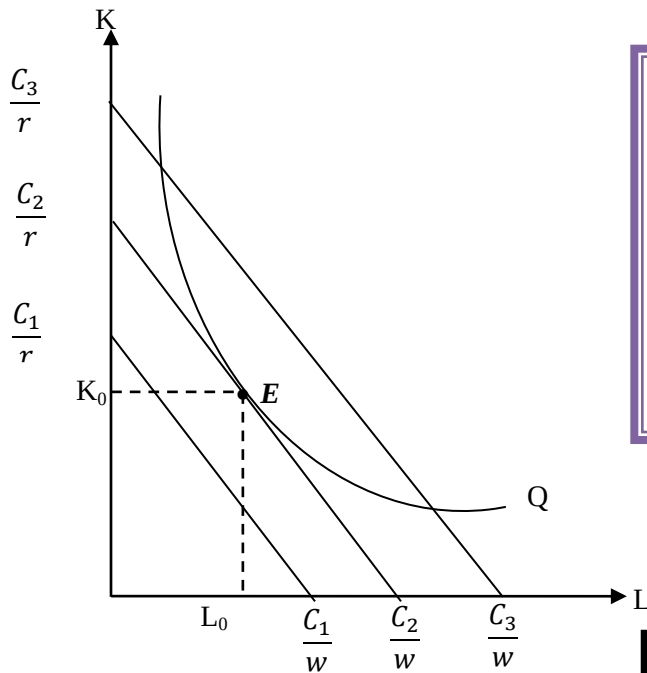
- ☞ In this case the total cost is known.

Fig 2.5 Output Maximization

B) Cost Minimization for a given level of output

The producer will be at equilibrium when he minimizes the cost of production for producing a specified level of output. This means that the producer reaches the lowest isocost for a particular output level (**Least cost rule**). This occurs where the isocost curve is just tangent to the isoquant curve (point “E”). At this point, the slope of the isoquant (MRTS) = slope of isocost line $\left(\frac{w}{r}\right)$.

i.e. $\frac{MP_L}{MP_K} = \frac{w}{r}$



Mathematically

Minimize $C = wL + rK \rightarrow$ Objective function

Subject to $Q = f(L, K) \rightarrow$ Constraint function

Then, the lagrangian function is:

$$L = wL + rK - \lambda[f(L, K) - Q]$$

☞ In this case the total output is known.

Fig 2.6 Cost Minimization

Numerical Example:

The production function for a firm is given by:

$$Q = 4L^{0.5}K^{0.5}$$

Suppose that the cost of labor is Birr 40 per unit and the cost of using capital is Birr 10 per unit.

- A) Determine the amount of labor and capital that should be used in order to minimize the cost to produce 40 units of output?
- B) Calculate the minimum cost of producing 40 units?

Solution :

A) Given: $Q = 4L^{0.5}K^{0.5}$, $w = 40$ birr per unit, $r = 10$ birr per unit, $Q = 40$ units

First, we should form the cost function: $C = wL + rK$ then, $C = 40L + 10K$

Minimize $C = 40L + 10K \rightarrow$ Objective function

Subject to $Q = 4L^{0.5}K^{0.5} \rightarrow$ Constraint function

Then, the lagrangian function ($\emptyset$) is:

$$\emptyset = 40L + 10K - \lambda(4L^{0.5}K^{0.5} - 40)$$

The first condition for the minimization of a function is that its partial derivatives be equal to zero. The partial derivatives of the above function with respect to L, K, and λ are:

$$\frac{\partial \emptyset}{\partial L} = 40 - \lambda 2L^{-0.5}K^{0.5} = 0 \rightarrow \lambda = \frac{40}{2L^{-0.5}K^{0.5}} \dots \dots \dots (1)$$

$$\frac{\partial \emptyset}{\partial K} = 10 - \lambda 2L^{0.5}K^{-0.5} = 0 \rightarrow \lambda = \frac{10}{2L^{0.5}K^{-0.5}} \dots \dots \dots (2)$$

$$\frac{\partial \emptyset}{\partial \lambda} = 4L^{0.5}K^{0.5} - 40 = 0 \rightarrow (3) \dots \dots \dots (3)$$

From (1) & (2): $\lambda = \lambda$

$$\frac{40}{2L^{-0.5}K^{0.5}} = \frac{10}{2L^{0.5}K^{-0.5}}$$

$$\frac{40}{10} = \frac{2L^{-0.5}K^{0.5}}{2L^{0.5}K^{-0.5}}$$

$$4 = \frac{K}{L} \rightarrow K = 4L$$

From the production function: $Q = 4L^{0.5}K^{0.5}$

$$40 = 4L^{0.5}(4L)^{0.5}$$

$$40 = 4L^{0.5}4^{0.5}L^{0.5}$$

$$40 = 4^{1.5}L$$

$$L = \frac{40}{8} = 5 \text{ units of labor}$$

Then, $K = 4L = 4(5) = 20$ *units of capital*

B) $C_{\text{minimum}} = 40L + 10K = 40(5) + 10(20) = 400$ **Birr** → *the minimum cost of production*

3.3.7. The Expansion Path in the Long-Run

The expansion path in the long-run is the expansion of output with all factor's variable. An expansion path is the locus of all equilibrium points when input prices remain constant and expenditure on inputs increases. The producer moves on higher level of output in this type of change. The expansion path is a useful concept because it gives an idea of how input proportion changes with increase in expenditure of the producer, input prices being constant. For a homogeneous production, the expansion path will be a straight line. Thus, the expansion path shows a producer's use of inputs will vary as the firm expands output.

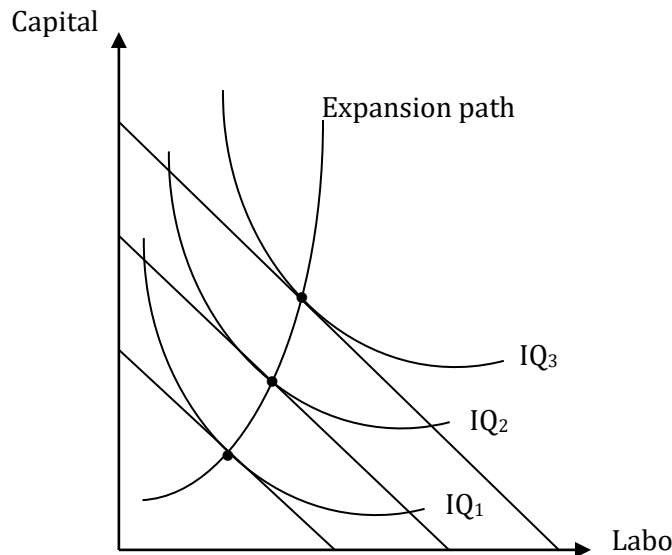


Fig 2.7 The Expansion Path

Assuming that the producer minimizes the cost of producing any given output, we can derive the expansion path by connecting point of tangency of isoquants and isocost lines. The minimum possible cost of producing each output is given by the tangency of each isoquant and isocost line.

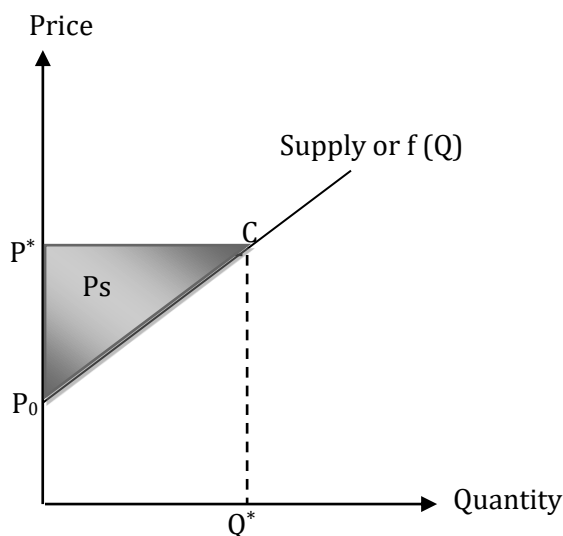
Notice the expansion path in fig 2.7 bends upward as output is increased by moving to higher and higher isoquants. The shape of the expansion path implies that the ratio of capital input to labor input is increase as output expands. The ratio of labor to capital would be constant only in the case of a straight-line expansion path.

The expansion path shown in fig 2.7 is quite realistic for most enterprises in the long-run because production methods become more capital- intensive as an enterprise grows. This is so, because as firms become larger, they tend to become more mechanized by increasing the ratio of capital to labor which is used to produce a given level of output.

3.4. Producer's Surplus (Ps)

Producer's surplus refers to the difference between what the producer actually receives over the minimum value and what he or she would have been willing to receive.

Geometrically, the producer surplus is measured by the area of the triangle between the supply curve and the market price. Given supply function: $P=f(Q)$, and equilibrium price and output (P_0, Q_0).



To find the shaded area or the producer surplus (Ps) we can use the triangle formula:

1

Fig 2.8 Producer's Surplus

Chapter-Two: Review Questions

Part I: Choose the best answer from the given alternatives

- 1) One of the following is **false** about Stage –II of production
 - A) It is the stage of diminishing returns because TP increase at decreasing rate.
 - B) MP_L is greater than AP_L .
 - C) Both MP_L & AP_L are diminishing but positive.
 - D) MP_L decline at slower rate than AP_L .
 - E) None

- Part II: Give the appropriate answer for the following questions (NB. show all the necessary steps for the workout part)**

- $$Q = 6KL^2$$

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- A) Determine the amount of labor and capital that should be used in order to minimize the cost to produce 1458 units of output?
- B) Calculate the minimum cost of producing 1458 units?
- C) Calculate MP_L and MP_K ?
- D) Mathematically state the returns to scale of the production process?
- 8) The production function for a firm is given by:

$$Q = KL^2$$

Suppose that labor costs Birr 20 per unit and capital costs Birr 30 per unit. Then,

- A) Determine the quantity of labor and capital that the firm uses in order to maximize the output with total outlay of Birr 450?
- B) Calculate maximum level of output?
- C) Calculate MP_L and MP_K ?
- D) Determine the factor intensity of the production process? And why?
- E) State the returns to scale of the production process?
- 9) Assume $Q = -L^3 + 15L^2 + 10L$, then find
- A) The maximum AP_L ? C) The maximum TP_L ?
- B) The maximum MP_L ? D) The stages of production?
- 10) Suppose Arba Minch textile factory is facing the following production function of fixed supply of machineries (capital) and variable input (labor).

Machineries (in number)	Labor (in work-hour)	TP	AP	MP	Stage of production
10	0	0			
10	1	2			
10	2	5			
10	3	9			
10	4	12			
10	5	14			
10	6	15			

10	7	15			
10	8	14			
10	9	12			

- Calculate the corresponding values of AP_L and MP_L ?
- Draw curves for TP, AP and MP of labor?
- Identify the three stages of production and show their tabular and graphical justifications?
- Indicate the point (unit level of labor) where the LDNR starts to operate?

CHAPTER FOUR

THEORY OF COST

In this unit, you will study the meaning and behaviors of costs of production, the relationship between production (output) and costs (i.e., cost function both in the short run and long run).

After successful completion of this unit, you will be able to:

- Explain different ways of measuring private costs, i.e., economic costs vs. accounting costs.
- Define the meaning and nature of cost functions both in the short run and long run.
- The distinction between short run and long run cost.
- Explain the relationship between short run production function and short run cost function.
- State how learning and experience affects the costs of production.
- The tradition theory of cost.

4.1. Introduction to Theory of Cost

The concepts of production and cost are inseparable facts. A production process in its pursue to secure profits will not exist without incurring the related costs. And the existence of costs without production or any addition to the total output will be of no economic meaning. Cost of production generally stands for the monetary outlays associated with production activity or it is the total expenditures and sacrifices made in the entire process of production and distribution of goods and services. In general, the cost of everything that is used up in the process of production is what we call the ***cost of production***.

Cost functions are derived functions. They are derived from the production function, which describes the available efficient methods of production at any one time.

Economic theory distinguishes between short-run and long-run costs. Short-run costs are the costs over a period during which some factors of production (usually capital equipment and management) are fixed. The long-run costs are costs over a period long enough to permit the change of all factors of production. In the long-run all factors become variable.

Both in the short-run and in the long-run, total cost is a multivariable function, that is, total cost is determined by many factors. Symbolically long-run cost function can be written as

$$C = f(X, T, P_f)$$

And the short-run cost function can be written as

$$C = f(X, T, P_f, K)$$

Where C = Total Cost, X = Output, T = Technology, P_f = Prices of factors, K = Fixed factor(s).

Based on the above facts, **costs** are defined as the monetary values of inputs used in production of items.

4.2. Types of Costs

A) *Explicit Vs Implicit costs*

Explicit costs: these are the **actual monetary payments** or **cash outlays** that business firms make to outsiders who are suppliers of inputs or resources to them. For example, the rewards of labor, land, capital are all costs for a business firm employing them in certain production process. In addition, there are other payments made for other raw materials, fuel, transport, power and the like are all costs to a firm. Such costs are usually termed as *accounting costs*. Thus, **accounting cost** refers to the cost of purchased inputs only and this is the explicit cost paid “out of pocket.”

Implicit costs: - costs standing for the **values** of **non-purchased resources** owned and used by firms in their own production activities are said to be *implicit costs*. There are costs of firms owned and self-employed resources in carrying activities like the salary of an owner-manager or the estimated rent of a building that belongs to the owner of a firm and etc. The values of these self-owned resources should be estimated from what they could earn in their best alternative use.

$\text{Explicit costs} + \text{Implicit costs} = \text{Economic costs (opportunity costs)}$

B) *Economic or Opportunity cost*

The opportunity cost of a good or service is the value of the best alternative that is forgone in order to produce a good or undertake the service. These costs are also called as ***economic costs*** for it includes both explicit costs (costs involving actual payments) and implicit costs (the benefits that would have resulted from choosing the sacrificed alternatives but do not involve direct payments).

Economic Vs accounting profits

It is clear that costs and profits are inseparable concepts of business. Here the main insight is to understand the cost treatment differences and its consequence in profit analysis of business activities.

The term profit refers to the difference between total revenue and total cost. But economists and accountants define it differently for they treat costs differently. Economists define costs in terms of opportunity costs and they include implicit costs in profit calculations. Thus,

$$\textbf{Economic profit} = \textbf{Total revenue} - (\textbf{implicit} + \textbf{explicit costs})$$

As the economic profit becomes zero, it means that the activity is earning the same profit as the next best alternative use of the employed resources because it is total revenue less opportunity cost.

Accounting profit, on the other hand, would be the difference b/n total revenue and accounting cost. Thus,

$$\textbf{Accounting profit} = \textbf{Total revenue} - \textbf{explicit costs}$$

C) Social Vs Private Costs

Social costs: are costs which a society bears on account of production of a commodity. Social costs include both private and the extra cost. Extra cost includes:

- i) The cost of resources for which the firm is not compelled to a price e.g. atmosphere, rivers, lakes, roads and drainage systems.
- ii) The cost in the form of disutility created through air & water and noise pollution etc.

Private costs: are costs which are actually incurred or provided for by an individual or a firm on the purchases of goods and services from the market. For a firm all the actual costs both explicit and implicit are private cost in the sense that the firm must compensate the resource owner in order to acquire the right to use the resource.

D) Fixed Vs Variable costs

Fixed costs (FC): are those costs that do not vary as the firm changes the level of output. These costs are costs that are always incurred even if the firm does not produce anything. These are also costs of fixed inputs. E.g., rents on leased properties, interest on borrowed funds, the wear and tear of machineries, cost of administrative staff and etc.

Variable costs (VC): are those costs of production that directly vary with the level of output of the firm. These are also costs of variable inputs. When output is zero variable costs are also zero. But as the firm expands its output these costs tend to rise. Therefore, can be expressed as a function of output (Q) i.e., $VC = f(Q)$

Based on the above explanations of types of costs please answer the following questions

- i) Consider a barber in your locality. What are the explicit and implicit costs that the barber incurs to provide the service?***
- ii) What are the main differences between accounting and economic ways of measuring costs? Which one do you believe is more appreciate? Why?***
- iii) Explain the distinction between social and private costs of production***

4.3. Theory of Cost in the Short-Run

Like short-run production, the short-run cost is a period during which there is at least one variable cost and at least one fixed cost in the process of production. Normally, the cost of labor considered as a variable cost and the cost of capital considered as a fixed cost. In the short-run, costs constitute both ‘variable costs’ and ‘fixed costs. The nature and behavior of different types of short run costs are examined and discussed below with the help of the following table and graphical presentation.

Table 3.1: Short-run costs (in Birr)

Quantity	TFC	TVC	TC	AFC	AVC	AC	MC
0	100	0	100	-	-	-	-
1	100	50	150	100	50	150	50
2	100	90	190	50	45	95	40
3	100	120	220	33.33	40	73.33	30
4	100	140	240	25	35	60	20
5	100	150	250	20	30	50	10
6	100	170	270	16.67	28.33	45	20
7	100	200	300	14.29	28.57	42.86	30
8	100	240	340	12.50	30	42.50	40
9	100	290	390	11.11	32.22	43.33	50
10	100	360	460	10	36	46	70

A) Total Fixed Cost (TFC): is constant through the period i.e., 100 Birr

B) Total Variable Cost (TVC): is zero when output is zero and rises as output rises but at varying rates. First at a decreasing rate and then at an increasing rate.

C) Total Cost (TC): is the aggregate money expenditure on the inputs used to produce a given level of output. It is 100 when output zero and after this it follows the same pattern as TVC but differ only in amount as total cost is the sum of TFC and TVC.

$TC = TFC + TVC$ But, in the long-run since there is no fixed cost, total cost is equal to total variable cost only (i.e., $TC=TVC$)

D) Average Fixed Cost (AFC): it is the ratio of TFC over total quantity (output). i.e., $AFC = \frac{TFC}{Q}$. TFC is constant, due to this reason AFC falls continuously as output increases.

E) Average Variable Cost (AVC): is the ratio of TVC over total quantity (output). i.e., $AVC = \frac{TVC}{Q}$ First it declines then reaches its minimum at 6 unit of output and then rises as output increases.

F) Average Cost (AC): this is the total cost per unit output and is calculated by dividing the total cost by quantity produced. This means, $AC = \frac{TC}{Q}$. It can also be separated in to two parts, like Average Variable Cost (AVC) and Average Fixed Cost (AFC).

$$\text{Thus, } AC = \frac{TFC}{Q} + \frac{TVC}{Q}; AFC = \frac{TFC}{Q} \& AVC = \frac{TVC}{Q} \text{ then, } AC = AFC + AVC$$

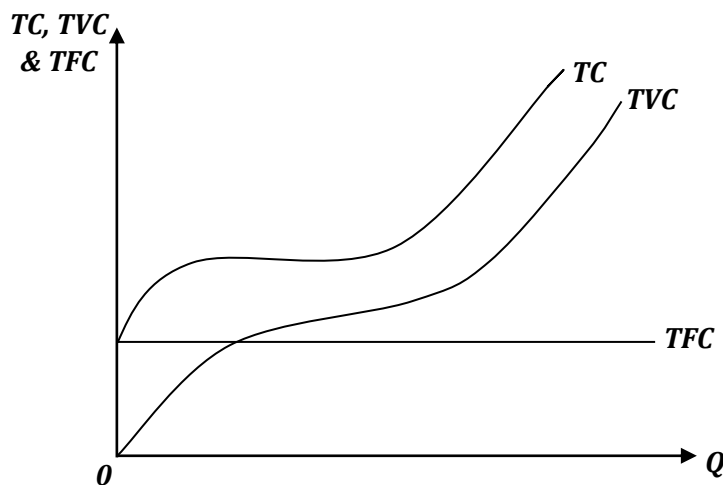
AC like AVC, initially decreases as quantity increases, reaches its minimum at 42.50 when the firm produces 8 units of output and then increases. Note that both AVC & AC curves have U- shape and AC reaches its minimum after AVC does.

G) Marginal Cost (MC): is the extra or additional total cost that results from producing one more unit of output or it is the change in total cost resulting from a percentage change in output i.e. $MC = \frac{\Delta TC}{\Delta Q}$ or $MC = \frac{\Delta TFC}{\Delta Q} + \frac{\Delta TVC}{\Delta Q}$

Or $MC = \frac{\Delta TVC}{\Delta Q}$, when $\Delta TFC = 0$ (in the short run)

It also initially falls, reaches its minimum at birr 10 when the firm produces 5 units of output and then rises. It reaches minimum before AVC & AC curves do. It is also a U-shaped curve. MC is the slope of TC curve (which is of course the same as the slope of the TVC)

Graphical presentation of short-run costs



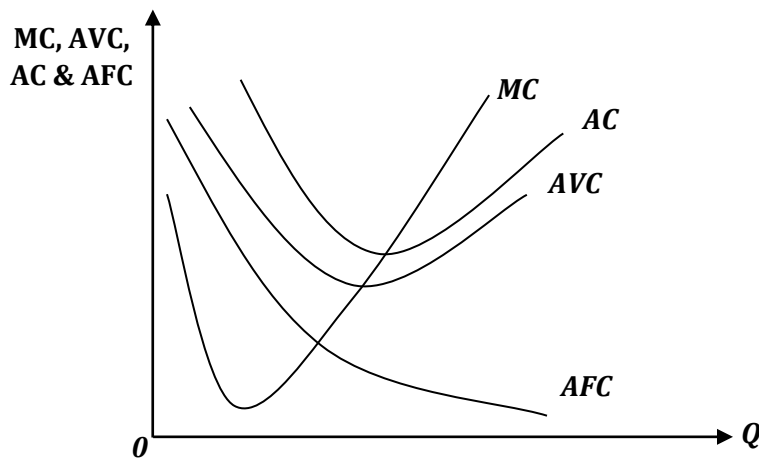


Fig 3.1 Short-run costs

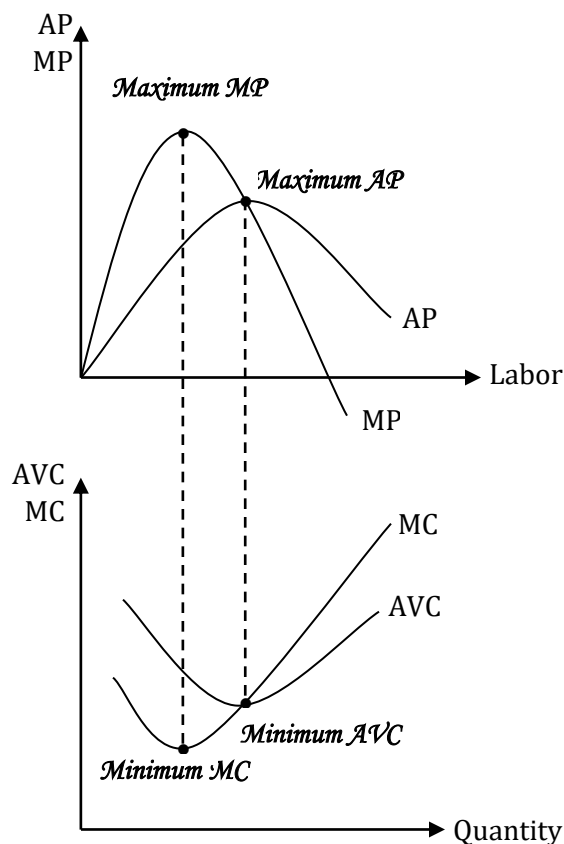
Now, note the following points from the diagram:

- ☞ For the output range where MC curve lies below the ATC curve, the ATC curve is falling. The ATC curve is rising for the output range where MC curve is above the ATC curve.
- ☞ For the output range where MC curve lies below the AVC curve, the AVC curve is falling. The AVC curve is rising for the output range where MC curve is above the AVC curve.
- ☞ The MC curve cuts both the ATC and AVC at their minimum points, while it is rising.
- ☞ The ATC is above the AVC for all output levels because the ATC is the sum of AVC&AFC.
- ☞ The gap between ATC and AVC is getting smaller and smaller for the reason that the AFC continuously decreases with the increasing output.
- ☞ The MC, AC, AVC curves are U-shaped due to the law of diminishing marginal returns.

*** The Relationship between Production and Cost**

There is an inverse relationship between AVC and AP: MC and MP. Now look at the following graph to understand the relationships in detail.

The MC curve and AVC curve are the mirror reflections of the MP and AP curves respectively.



The link between Production and Cost:

- ✓ When MP increase, MC decrease
- ✓ When MP decrease, MC increase
- ✓ When MP at its maximum, MC at its minimum

- ✓ When AP increase, AVC decrease
- ✓ When AP decrease, AVC increase
- ✓ When AP at its maximum, AVC at its minimum

- ✓ When $MC=AVC$, $MP=AP$

Therefore,

- ✓ MP is the mirror of MC
- ✓ AP is the mirror of AVC

Fig 3.2 The Link between Production and Cost

For the output level where MP is above AP, the MC is below AVC. For the output level where MP is below AP, the MC is above AVC. And for the output level where MP is equal to AP, the MC is equal to AVC. MP is equal to AP when AP is at its maximum point and hence MC equals AVC when AVC is at its minimum.

4.4. Theory of Cost in the Long-Run

The fundamental difference between the short-run and the long-run cost is that, in the short-run some costs are fixed where as in the long-run all costs become variable. It implies that in the long-run firms can hire more of both labor and capital.

In the long-run cost firm can control output and costs not only by varying the intensity of operating of a given plant but also by varying the size and the number of plants. Here the relationship is between changing size of the firm and outputs.

In the long-run there is no fixed input, therefore we didn't give attention to fixed cost. So, the total cost is equal to total variable cost and *the average cost is*: $AC = \frac{TC}{Q} = \frac{TVC}{Q} = AVC$

Since long-run is the sum of short-run, long-run cost curves would be composed of series of short-run cost curves. Now we shall derive long-run cost curves and study the relationship between cost and output.

Long –Run Average Cost (LAC) Curve

The derivation of the LAC has been lustrated in fig 4.3 below. As it is shown in the figure, there are three SAC (short-run average cost) curves corresponding to the three different plant size (small, medium, large size) each particular plant size generates its own short-run average total cost curve i.e. small size: SAC₁, medium size: SAC₂, and large size: SAC₃.

The AC of the first plant minimum cost is Q₁C₁. The AC of the second plant is lower than that of the first and its minimum cost is Q₂C₂ which is lower than Q₁C₁ due to economies of scale since the second plant is added (i.e. plant size increased). This happens because the larger plant size makes possible specialization and the availability of capital per worker increases and productivity of labor increases. However, when the third plant is brought in, diseconomies of scale happened because management becomes highly complex, managers are overworked and decision-making process becomes less efficient.

The long-run average cost curve can now be drawn by drawn by joining the falling part of the SAC curves for points to the left of the minimum point of the SAC₂; the rising part of the SAC curves for points to the right of the minimum points of the SAC₂, and at minimum of SAC₂ curve. This long-run average cost curve is also called '*Envelope*'/ '*Planning*' curve.

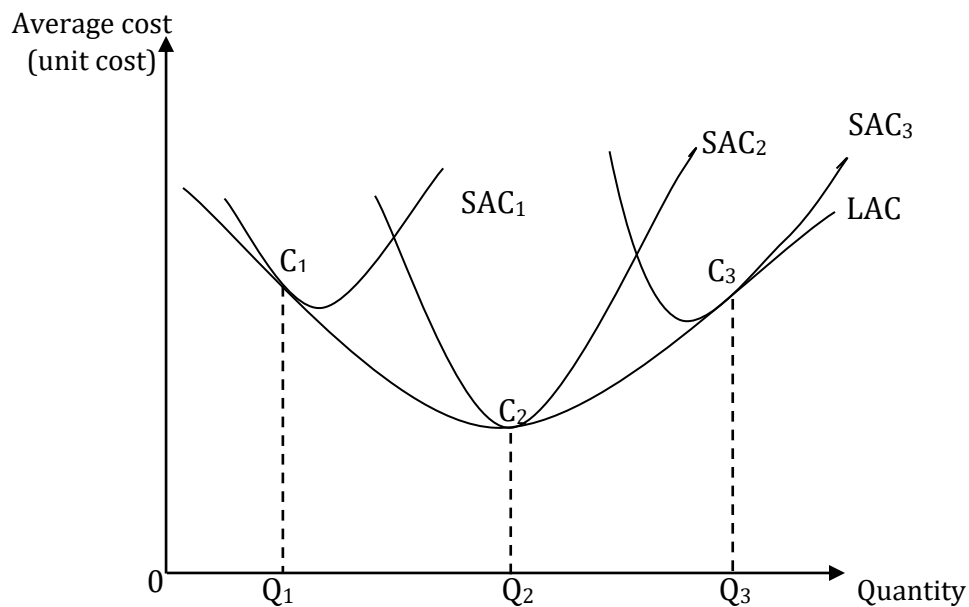


Fig 3.3 Derivation of Long-run Average Cost

As you can see in the figure 3.3, output Q_2 is the optimum as at this point LAC is minimum (which equals with the second plant i.e. medium size plant with minimum short-run average cost). This is because; in this plant all possible economies of scales are fully exploited. If the plant increase further than this optimum size (i.e. large size plant with SAC₃) diseconomies of scale will be created.

Chapter- Three: Review Questions

Part I: Choose the best answer from the given alternatives

- 1) Which one of the following is an implicit cost?
 - A) The price of a plane
 - B) Ticket to a football match
 - C) Time spent in line buying tickets
 - D) Rent payment for an apartment
 - E) None of the above

Consider a firm has the following costs for the short-run production of different quantities of a good:

Quantity produced(Q) (in units)	0	1	20	40	60	80	100
Total cost (C) (in Birr)	27	30	40	60	80	130	220

Answer questions 2-4 based on the above table:

- 2) The marginal cost of producing 80 units is,
 A) 1.2875 B) 1.625 C) 2.5 D) 4.5 E) None of the above
- 3) The fixed cost of production is,
 A) 30 B) 3 C) 220 D) 27 E) None of the above
- 4) The average variable cost when 40 units are produced is,
 A) 0.725 B) 1.5 C) 33 D) 0.825 E) None of the above
- 5) Which of the following is **false**?
 A) ATC always above the AVC
 B) With the increasing output, ATC & AVC are getting closer.
 C) TC change only due to the change in TVC
 D) AFC forming a rectangular hyperbola
 E) None of the above
- 6) The value of resources that belong to the owner of the firm that are employed in the production but they are not included in the cash flow of the company are:
 A) Private cost D) Accounting cost
 B) Social cost E) Implicit cost
 C) Economic cost
- 7) All the following are **true except one** if the Short run MC curve equal to short run AC curve, equal to long run MC curve, and equal to long run AC curve
 A) Long run full capacity production
 B) The unit cost of production is minimum
 C) The firm have an economies of scale advantage
 D) The inflection point of economies of advantage
 E) None of the above

Part II: Give the appropriate answer for the following questions

- 1) Analyze the relationship between production and cost using the relevant tool of analysis?
- 2) As a matter of cumulative experience with production the firm's marginal and average costs of producing a given output decreases because of certain reasons. Identify and give description of four reasons that contributes for the above-mentioned idea.
- 3) Explain the scenarios behind the "U" shaped curves of the traditional short-run curves of cost.
- 4) Consider a barber in your locality. What are the explicit and implicit costs that the barber incurs to provide the service?
- 5) Discuss how long run average cost change due to economic of scale and learning scale.
- 6) Assume the total cost function is: $TC = 10 + 6Q - 0.9Q^2 + 0.05Q^3$, then find
 - A) The AFC & AVC function?
 - B) The Minimum AVC?
 - C) The Minimum AC?
 - D) The Minimum MC?

CHAPTER FIVE

PERFECT COMPETITION

In this chapter, we shall try to see how a given firm operating in a perfectly competitive market determines the profit maximizing level of output and price, and how equilibrium market price and level of output are determined in a perfectly competitive market. Our discussion starts with giving a brief description about perfect competition.

After successful completion of this unit, you will be able to:

- Characterize a perfectly competitive market.
- Know how a perfectly competitive firm determines the profit maximizing output both in the long run and short run.
- Derive the short run supply schedule of an individual firm and industry.
- Explain when a perfectly competitive firm should decide to shut down.
- Explain how a perfect competition results in efficient allocation of resources.

4.1. Introduction: Definition and Basic Assumptions of Perfect Competition

Perfect Competition is a market structure characterized by a complete absence of rivalry among the individual firms as there are so many firms in the industry so that no personal recognition among the individual firms in the market.

Basic Assumptions of perfectly competitive market

Perfectly competitive market is characterized by the following assumptions:

A) Large number of sellers and buyers

The industry or market includes a large number of firms; so that on the supply side, each firm has small share relative to the total market, in that it cannot affect market price by changing its output supplied. The buyers are also numerous; so that on the demand side, individuals influence on the total market is very small. This means that each buyer purchases a small portion of the total market demand that he cannot obtain special consideration of price reduction from the seller.

B) Homogeneous product

The industry is defined as a group of firms producing a homogeneous (standardized) product. The product of each seller is identical to the product of every other seller. There is no way in which a buyer could differentiate among the products of different firms. If the products were differentiated the firm would have some discretion in setting its price. The assumption of large number of sellers and product homogeneity imply that the individual firm in pure competition is a **price-taker**. Its demand curve is infinitely elastic, indicating that the firm can sell any amount of output at the prevailing market price. The demand curve of the individual firm is also its average revenue and its marginal revenue curve

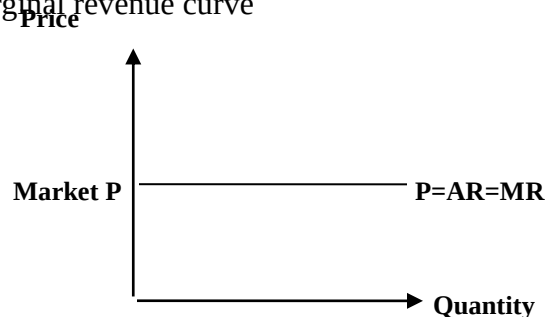


Fig 4.1 Firm's demand curve

C) Perfect knowledge (information)

It is assumed that all sellers and buyers have complete knowledge of the conditions of the market. Each buyer and seller have full information about price, quality of the product and other characteristics of the product. Information is free and costless. This knowledge refers not only to the prevailing conditions in the current period but in all future periods as well. Under these conditions uncertainty about future developments in the market is ruled out.

D) Free entry and exit of firms

There is no legal or market barrier to entry or exit from the industry. Firms have freedom of movement in and out of the industry.

E) No government regulation

There is no government intervention in the market (tariffs, subsidies, rationing of production and so on are ruled out).

F) Perfect mobility of factors of production

The factors of production are free to move from one firm to another throughout the economy. It is also assumed that workers can move between different jobs, which imply that skills can be learned easily. Finally, raw materials and other factors are not monopolized and labor is not unionized. In short, there is perfect competition in the markets of factors of production. In addition to this, there is no collusion among the buyers or sellers. The trade and manufacturing associations do not exist in the market.

Based on the above concepts please answer the following questions

- i) Is promotion activity (such as advertising the product) necessary for a firm operating under perfect competition? Why or why not?***
- ii) What are the assumptions used to construct the model of perfect competition?***
- iii) Explain why the demand curve of an individual firm operating in a perfectly competitive market is perfectly elastic. What will happen to the quantity of sales of an individual firm if he raises the price above the market price?***

4.2. Short-Run Equilibrium of the Firm and the Industry

In order to determine the equilibrium of the industry we need to derive the market supply. This requires the determination of the supply of the individual firms, since the market supply is the sum of the supply of all the firms in the industry.

4.2.1. Short-Run Equilibrium of the Firm

Profit maximization is the popular objective of business firms. Thus, the firm is said to be in equilibrium when it realizes maximum profit or minimum losses. There are two approaches to determine the profit maximizing (equilibrium) level of output; ***Total approach and marginal approach.***

A) Total Approach

In this approach the firm is in equilibrium when it maximizes its profit (π), defined as the difference between total cost (TC) and total revenue (TR). $\Pi = TR - TC$

Figure 4.2 shows the total revenue and total cost curves of the firm in a perfectly competitive market. The total revenue curve is a straight line through the origin; showing that the price is constant at all levels of output. Because the firm is price taker & can sell any amount of output at the ongoing market price, with its TR increases proportionately with its sales. The slope of the TR curve is the marginal revenue. It is constant and is equal to the prevailing market price, since all units are sold at the same price. Thus, in pure competition; $MR=AR=P$

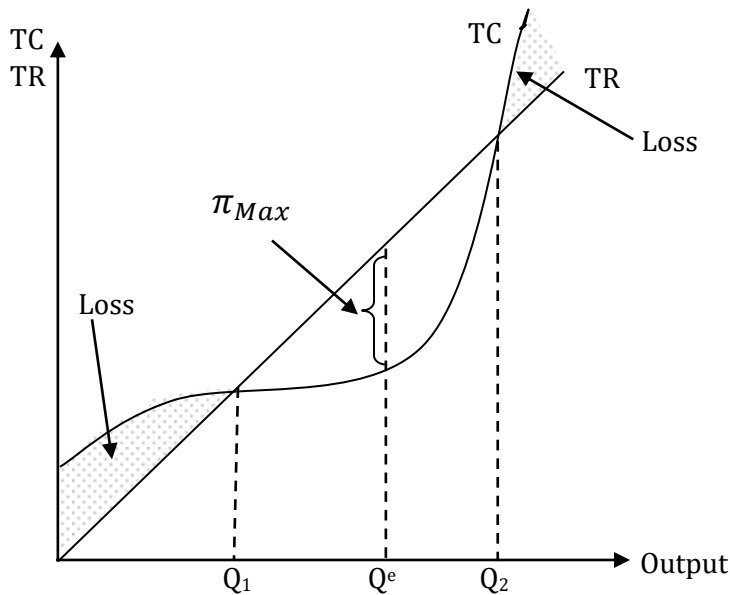


Fig 4.2 Total Approach

The firm maximizes its profit at the output Q^e , where the distance between the total revenue & total cost curves is the highest. At output levels smaller than Q_1 and larger than Q_2 the firm has losses.

B) Marginal Approach

When we use the marginal approach, two conditions should fulfill. The first order condition for maximization of a function is that its first derivative would be equal to zero.

$$\text{i.e. } \frac{d\pi}{dQ} = \frac{dTR}{dQ} - \frac{dTC}{dQ} = 0$$

$$\text{i.e. } \frac{dTR}{dQ} = \frac{dTC}{dQ} \rightarrow MR = MC \text{ (Slope of TR \& TC, respectively)}$$

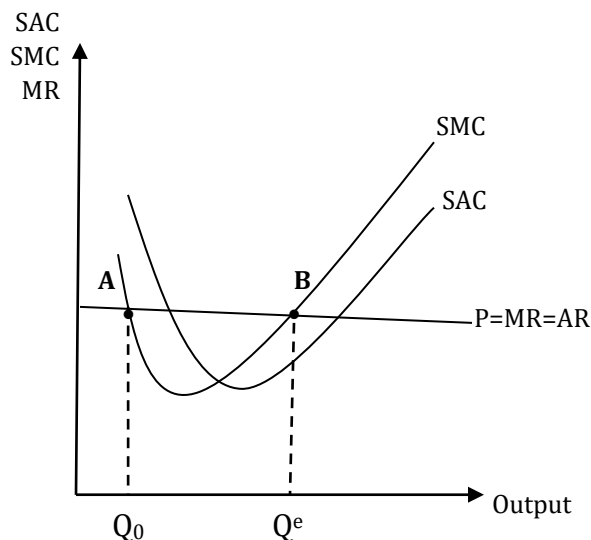
The second order condition for maximization of profit requires that the second order derivative of the function must be negative.

$$\text{i.e. } \frac{d^2\pi}{dQ^2} = \frac{d^2TR}{dQ^2} - \frac{d^2TC}{dQ^2} < 0$$

$$\text{i.e. } \frac{d^2TR}{dQ^2} < \frac{d^2TC}{dQ^2} \rightarrow \text{Slope of MR} < \text{slope of MC.}$$

As you know slope of the MR=0 in the perfect competition since it has horizontal demand curve. It should be noted that the MC positive. In summary the two conditions are:

- i) MC=MR
- ii) (Slope of MC) > (Slope of MR) → MC is rising so that it intersects the MR curve from below.

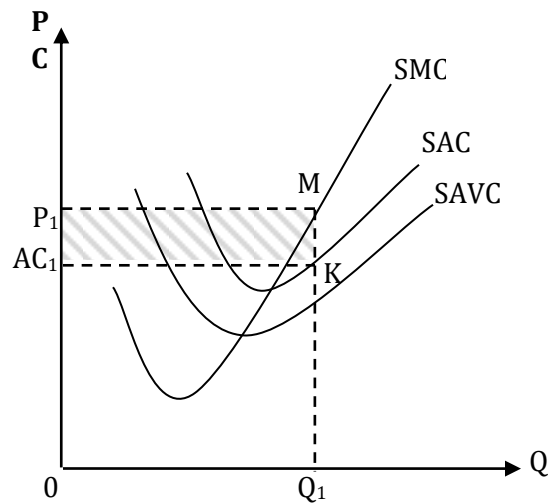


Therefore, Q^e is the profit maximizing output level.

Fig 4.3 Marginal Approach

❖ Now let us go through the following five different cases of the optimum price-output combinations.

- 1) **When $P > AC$ at the optimal level:** (E.g. P_1), revenue per unit (Average revenue = P_1) is greater than cost per unit (AC_1). As a result, the firm gets **Abnormal / Positive profit or Super-normal profit**.



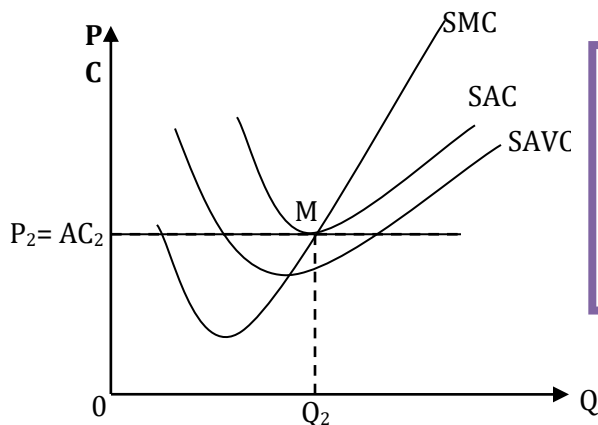
$$TR_1 = P_1 \cdot Q_1 = \text{area}(P_1MQ_1O)$$

$$TC_1 = AC_1 \cdot Q_1 = \text{area}(AC_1KQ_1O)$$

$$\text{Profit} = TR_1 - TC_1 = (P_1MKAC_1)$$

Fig 4.4

- 2) **When $P = AC$ at the optimal level:** (E.g. P_2), revenue per unit (Average revenue = P_2) is equal to cost per unit (AC_2). As a result, the firm gets **normal profit (zero economic profit)**. This is the case when the firm covers all of the economic costs of production, i.e. it covers both implicit and explicit costs; this point is also called **break – even point**, and P_2 is called the break – even price and Q_2 the break – even quantity.



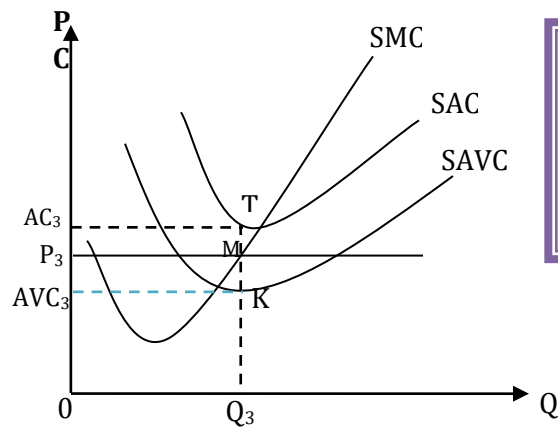
$$TR_2 = P_2Q_2 = \text{area}(P_2MQ_2O)$$

$$TC_2 = AC_2 \cdot Q_2 = \text{area}(AC_2MQ_2O) = \text{area}(P_2MQ_2O)$$

$$\text{Profit} = TR_2 - TC_2 = (P_2MQ_2O) - (P_2MQ_2O) = 0$$

Fig 4.5

- 3) **When the prevailing price is between the average cost and the average variable cost at the optimum level ($AVC < P < AC$),** the firm is said to be minimizing loss, because without production the firm has to incur the fixed cost amount but by producing the optimal output (Q_3) the firm loses less than the total fixed cost. In this case, the firm can cover a portion of its **total fixed cost** by producing the optimal output.



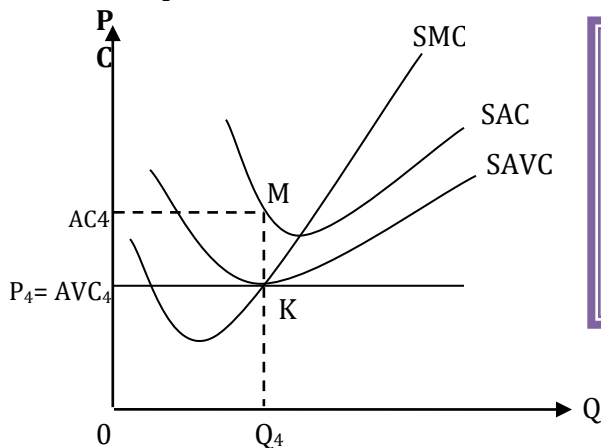
$$TR_3 = P_3 \cdot Q_3 = \text{area } (P_3 M Q_3 O)$$

$$TC_3 = AC_3 \cdot Q_3 = \text{area } (AC_3 T Q_3 O)$$

$$\text{Profit} = TR_3 - TC_3 = \text{area } (-AC_3 T M P_3) \rightarrow (\text{Loss})$$

Fig 4.6

- 4) When the price (P_4) is determined by the forces of market demand and supply is equal to the minimum average variable cost ($P = \min.AVC$), at that level of output, the firm is **indifferent between producing and not producing** because what the firm incurs by producing the optimum level and shutting business is the same (that is, the amount of fixed costs). Hence, we call this point the **shutdownpoint** and the corresponding price and quantity are called the shut down price and quantity, respectively. Still we advise the firm to continue production for the reasons of **learning by doing**, i.e. the firm may through time be able to reduce the costs per unit.



$$TR_4 = P_4 \cdot Q_4 = (P_4 K Q_4 O)$$

$$TC_4 = AC_4 \cdot Q_4 = (AC_4 M Q_4 O)$$

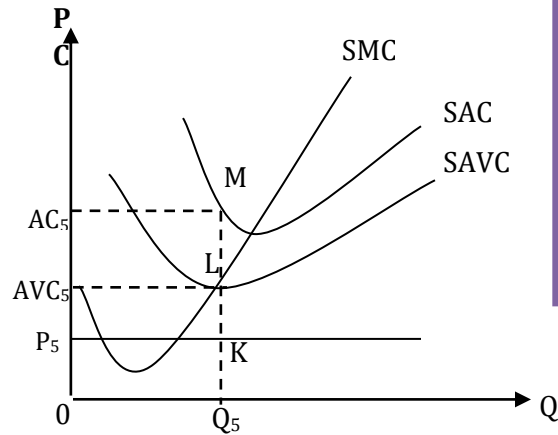
$$\begin{aligned} \text{Profit} &= TR_4 - TC_4 = -AC_4 M K P_4 \rightarrow (\text{Loss}) \\ &= -AFC \cdot Q_4 = -TFC \end{aligned}$$

Fig 4.7

☞ In this case, the firm is able to cover fully its variable cost, but not its fixed costs.

- 5) Finally, when the price is below the minimum point of the AVC ($P < \min.AVC$) curve at the “optimum level”, we say the firm exits out of the market because the firm loses more than the

total fixed costs if it decides to operate. Therefore, the firm incurs more by producing than shutting its business. Let's see the outcome of producing at the "optimal level".



$$TR_5 = P_5 \cdot Q_5 = (P_5 K Q_5 O)$$

$$TC_5 = AC_5 Q_5 = (AC_5 M Q_5 O)$$

$$\text{Profit} = TR_5 - TC_5 = -AC_5 M K P_5 > TFC \text{ by } (AVC_5 L K P_5)$$

Fig 4.8

☞ In this case, it is more economical to exit out of the market in the short run in order to save $(AVC_5 L K P_5)$ amount.

Numerical Example:

If the total cost function of a firm under perfectly competitive market is given by:

$$TC = 2Q^2 - 28Q + 100$$

(a) Find the optimum level of output and the corresponding profit when price of the product is Br. 20?

Solution: $MC = 4Q - 28$

$$P = MR = 20 \text{ Br.}$$

$$MC = MR = P \rightarrow 4Q - 28 = 20$$

$$\therefore Q^* = 12 \text{ units}$$

$$\Pi_{12} = TR - TC = 240 - 52 = 188 \text{ (maximum profit), where } TR = PQ$$

(b) If the total fixed cost has increased by Br. 100 and the price of the product is reduced to Br. 4 per unit. Then, what is the optimum production level and the firm's profit? If you are the management advisor for the firm, what do you suggest?

Answers $Q^* = 8$

$$TC = 2Q^2 - 28Q + 200 = 104 \text{ and } TR = 4 \times 8 = 32 \text{ then } \Pi = TR - TC = 32 - 104 = -72$$

⇒ Continue production in order to minimize loss, because loss is less than TFC (i.e. $72 < 200$)

Derivation of Short-run Supply Curve of the Firm

Dear distance learner! The supply curve of an individual firm is derived on the basis of its equilibrium output. The equilibrium output which is determined by the intersection of MR & MC curve is the optimum supply by a profit maximizing firm. To remain in the business and continue production, the firm must cover its SAVC. For this reason, the price must be at least equal to SAVC i.e. P_1 . As shown in the figure below, the firm produces and supply different level of output with their respective price. But if the price decreases further to p_1 , the firm stops production and its supply will be zero. From this we can understand that the firm's supply curve is that the segment of the MC curve above the minimum of SAVC.

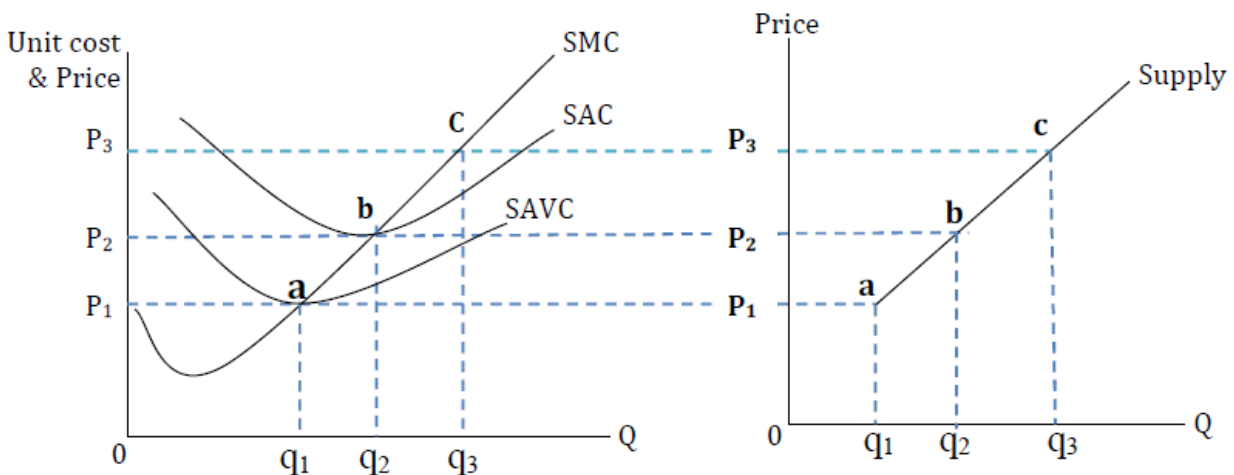


Fig 4.9 Derivation of Firms Supply Curve

Note, however, that this profit or loss making is a short run phenomenon where as in the long run all the firms earn only normal profit because a super normal profit will attract new entrants in to the market, substantially affecting the market supply and consequently the price. On the other hand, a firm will not operate making a loss since there is free exit out of the market.

4.2.2. Short-Run Equilibrium of the Industry

Given the market demand and the market supply, the industry is in equilibrium at that price which clears the market; i.e. at the price at which the quantity demanded is equal to the quantity supplied. In fig.4.10 the industry is in equilibrium at price P^e , at which the quantity demanded and supplied is Q^e . However, this will be short run equilibrium, if at the prevailing price firms are making excess profit: $P > C$ 4.11 or losses Fig. 4.12. In the long run, firms that make losses and cannot readjust their will close down. Those that make excess profits will expand their capacity, while excess profits will also attract new firms into the industry. Entry, exit and readjustment of the remaining firms in the industry will lead to a long-run equilibrium in which firms will just be earning normal profits and there will be no entry or exit from the industry.

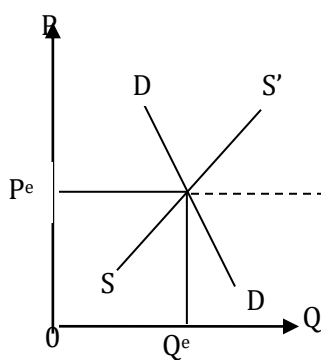


Fig 4.10

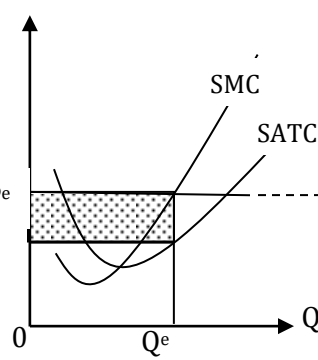


Fig 4.11

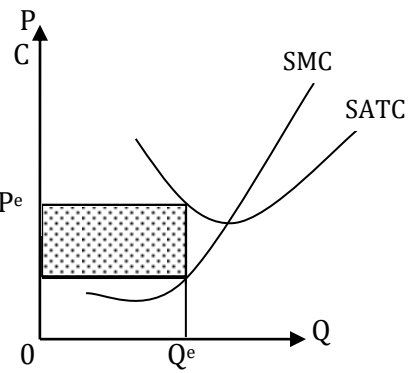


Fig 4.12

The supply of the industry in the short-run is the sum of the supply of the firm within the industry.

4.3. Long-Run Equilibrium of the Firm and the Industry

4.3.1. Long- Run Equilibrium of the Firm

The short-run by definition, is a period in which firms cost and revenue curves are given, firms cannot change their size as capital is fixed, existing firms cannot leave the industry, and new firms cannot enter the industry. By contrast, long run is a period in which these constraints are absent. Long-run permits change in technology and employment of both labor and capital i.e., firms can change their size, some of the existing firms may leave and new firms may enter the industry. In the long run supply curve not only shift right ward but also becomes more elastic.

Thus, a firm is in equilibrium under perfect competition when marginal cost is equal to price (MR). But for the firm to be in the long run equilibrium, besides MC being equal to AC. This is because, if the Price is greater than average cost, the firm will earn more than normal profit. These super normal profits will attract other firms in to the industry. With the entry of new firms in the industry, the price of the product will decline as the result of the increase in the supply of the output and the cost will increase as a result of more competition for factors of production as the industry expands. The firms will enter to the industry until the price is equal to long run average cost (LAC) so that all firms are earning only normal profit. If the price is below the average cost the firms in the industry incur losses. To avoid heavy losses, some firms leave the industry. This decreases the production and forces the price to stay at the average cost, so that the firms remaining in the industry earn normal profit.

It, therefore, follows that a perfectly competitive firm to be in the long-run equilibrium, the following conditions must be fulfilled.

1. Price (AR)= LMC=MR
2. Price =Average cost

Hence, in the long-run, the firm will be in equilibrium if; $P=AR=LMC=LAC=MR$, this condition can be illustrated with the help of the following graph.

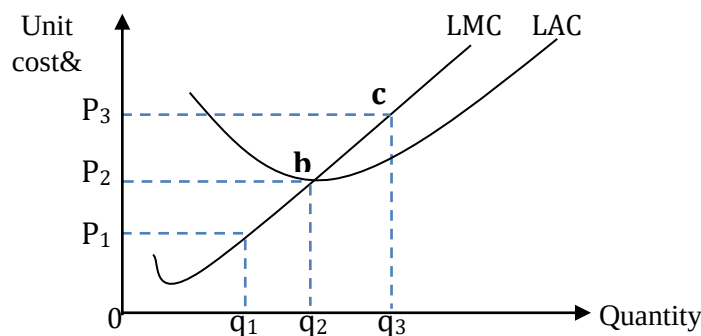


Fig 4.13 Long-run Equilibrium of the Firm

At price OP_3 , the firm is in equilibrium at point 'c' at output Oq_3 . The firm earns per unit abnormal profit of 'cb', as a result new firms enter the industry and increase production. The increase in production lowers down the price to OP_2 . If the price is OP_1 which is below LAC, firms will incur losses. In consequence, to avoid losses, the firms will leave the industry in the

long run. This reduce supply in the market raising the price to $0P_2$ & equal LAC; where the firm makes normal profit and the LAC is tangent to the demand curve ($P_2 = MR_2 = AR_2 = DD_2$). So, in the long run firm's supply will be the output level in which:

$$P = AR = MR = LMC = SMC = LAC = SAC$$

4.3.2. Long- Run Equilibrium of the Industry

The industry is in the long run equilibrium when a price is reached at which all firms are in equilibrium (producing at the minimum point of their LAC curve and making just normal profits). Under these conditions, there is no further entry or exit of firms in the industry.

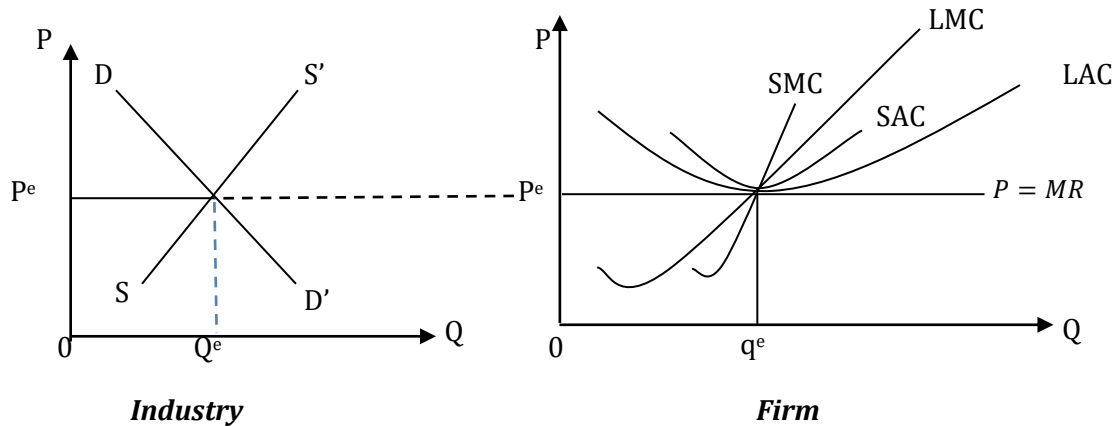


Fig 4.14 Long-run Equilibrium of the Industry

At the price P^e the industry is in equilibrium because profits are normal and all costs are covered so that there is no incentive for entry or exit. That the firms earn just normal profit (neither excess profit nor losses) is shown by the equality

$$P = MR = LMC = SMC = LAC = SAC$$

This is observed at the minimum point of the LAC curve. With all firms in the industry being in equilibrium and with no entry or exit, the industry supply remains stable, and, given the market demand (DD' in figure 5.14), the price P^e is a long-run equilibrium price.

Chapter-Five: Review Questions

Part I: Choose the best answer from the given alternatives

- 1) One of the following is **true** about short-run equilibrium of the perfect competitive firm
 - A) The second order derivative would be equal to zero at equilibrium
 - B) Slope of MC is less than the slope of MR
 - C) Total revenue is a convex to the origin
 - D) At equilibrium the firm will maximize its losses
 - E) None of the Above
- 2) Assume a profit-maximizing firm in perfect competition industries producing soft. The firm sells its soft at the prevailing market price, why? Due to the assumption of
 - A) Large number of sellers and buyers, and producing similar product
 - B) Producing homogenous product and no individuals influence on the market
 - C) Cost less information and no government regulation
 - D) Free entry and exit of firms
 - E) Perfect mobility of factors of production
- 3) **Except one**, all of the following are **true** about the long-run equilibrium of perfectly competitive firms
 - A) Firms will leave the industry to avoid normal profits
 - B) Long-run average cost curve is tangent to the demand curve
 - C) Firms can change their plant size
 - D) Firms earn a normal profit
 - E) Price and average cost of the firm are equal due to free entry
- 4) One of the following is **not** assumption of perfectly competitive market
 - A) There is a personal recognition
 - B) Single buyer and sellers cannot affect the market price
 - C) Any amount of output sells at constant price
 - D) Consumers have no any room to differentiate the products of different firms
 - E) Government doesn't intervene in the market

- 5) Firms within perfectly competitive market may encounter one of the following in short run depending on average cost:
- A) Excess profit
 - B) Normal profit
 - C) Negative profit
 - D) All of the above
 - E) Unknown
- 6) In a perfectly competitive market, which one of the following is **false** when the prevailing price is between the average cost and the average variable cost at the optimum level in the short-run?
- A) The firm loses less than the total fixed cost
 - B) The firm can cover some portions of its total fixed cost
 - C) It is more economical to exit out of the market
 - D) The firm can cover all of its total variable cost
 - E) Without production the firm incur the fixed cost
- 7) Suppose a profit-maximizing firm in a competitive market produces gasoline. Which one of the following is **false** when the market price for gasoline falls below the minimum of its average total cost, but still lies above the minimum of average variable cost?
- A) The firm loses less than the total fixed cost
 - B) The firm can cover a portions of its total fixed cost
 - C) It is more economical to exit out of the market
 - D) The firm can cover all of its variable costs
 - E) The firm is said to be minimizing loss
- 8) In a perfectly competitive market, which one of the following is **true** when the prevailing price is at the minimum point of the average variable cost in the short-run?
- A) It is more economical to exit out of the market
 - B) The firm can cover some portions of its total fixed cost

- C) The firm can cover all of its total variable cost
 - D) The firm loses is equal to the total fixed cost
 - E) It is a break-even point of production
- 9) In short run, supply curve of perfectly competitive firm is
- A) The segment of MC curve above its minimum point
 - B) The segment of AVC curve above its minimum point
 - C) The segment of MC curve above the minimum point of AVC curve
 - D) The segment of AC curve above its minimum point
 - E) The segment of AC curve above the minimum point of MC curve

Part II: Work out -Show all the necessary steps

- 1) If the total cost function of a firm under perfectly competitive market is given by:

$$TC = 3Q^2 + Q + 90. \text{ Then,}$$

- a) Find the optimum level of output and the corresponding profit when the price of the product is Birr 25? (3 pts)
- b) If you are the advisor of the firm, what do you suggest? And why? (2 pts)

CHAPTER SIX

PURE MONOPOLY

Introduction

In this chapter we would like to instigate you with some of the basic concepts and principles in the pure monopoly market structure. We start with monopoly market definition, discussing its characteristics and broaden our scope on gradual basis towards the discussion of short run and long run equilibrium of the market. But we need to keep our mind cool as to the depth of our discussion in this chapter as well.

Chapter objectives: Generally speaking, the learner will be expected to get familiar with the following key issues:

- What a monopoly market is all about;
- The main marketing principles that are accomplished while working on monopoly market and
- Social cost of monopoly market

6.1. Characteristics and Source of Monopoly

Definition: A monopoly is a market that has only one seller, but many buyers. A monopoly is the polar opposite of perfect competition. As the sole producer of a product, a monopolist is in a unique position. In a perfectly competitive market, there are enough sellers and buyers of a good so that no single seller or buyer can affect its price. Price is determined by the market forces of supply and demand. Individual firms take the market price as a given in deciding how much to buy. If the monopolist decides to raise the price of the product, it need not worry about competitors who, by charging a lower price, would capture a larger share of the market at the monopolist's expense. The monopolist is the market and has complete control over the amount of output offered for sale.

The following points are worth mentioning while discussing on the inherent characteristics of a monopoly market structure in a given area.

Characteristics of Monopoly

The following points are common and worth mentioning while discussing on the inherent characteristics of a monopoly market structure in a given area.

- i. **Single Seller:** It is a market structure in which the entire supply is controlled by one firm, which implies that the firm and industry are same.
- ii. **No clear substitutes:** there is no any close substitutes for the goods produced.
- iii. **The monopolist is price maker:** It does not a price, which is determined by the interaction of market demand and supply. In order to expand its sale, it decreases the price of a product.
- iv. **Entry Barrier:** Entry is blocked in such market structure. The barriers may be legal, financial, and natural.
- v. **No Collusion and Competition:** Because there is only one firm there is no competition exists and no collusion among firms also.

Causes of Monopoly:

- 1. Exclusive knowledge of technology.
- 2. Exclusive ownership/access to strategic raw materials needed for production.
- 3. Government Policies: Patent Rights.
- 4. Size of market:
- 5. Natural Monopolies: Electricity, Water etc.,

Dear student, given the above characteristics of a monopoly market structure, let you reflect on your views on the following lines.

❓ Define the monopoly market structure as a working concept in your day-to-day life. It would be appreciated if you substantiate your definition with real world examples.

❓ What do you think is the basic difference between perfect competition and monopoly market structures [in the context of microeconomics]?

Demand and Revenue for a Monopoly

The whole market supply is controlled by a single firm. In order to sell more output, the firm reduces price and to increase price the firm reduces its output. Consequently, output and price decisions are interdependent. So that increasing of one will result in a reduction of the other and vice versa. As a result, the demand curve of the monopoly is downward sloping.

Note: The monopoly firm faces the industry demand function which is negatively sloped.

Suppose the demand function of the monopoly is given by $Q = a - bP$.

$$\begin{aligned} \text{Solve for } p \quad P &= \frac{a}{b} - \frac{1}{b}Q \\ TR &= PQ \\ TR &= \left[\frac{a}{b} - \frac{1}{b}Q \right] Q \Rightarrow TR = \frac{a}{b}Q - \frac{1}{b}Q^2 \\ AR &= \frac{TR}{Q} \\ AR &= \frac{a}{b} - \frac{1}{b}Q \end{aligned}$$

At any level of output average revenue and price are the same.

$$MR = \frac{dTR}{dQ} = \frac{d\left[\frac{a}{b}Q - \frac{1}{b}Q^2\right]}{dQ} = \frac{a}{b} - \frac{2}{b}Q$$

Comparison of price and marginal revenue will result in having identical vertical intercepts, but marginal revenue is twice as steeper as the inverse demand.

$$\text{From } TR = PQ \text{ and From } MR = \frac{dTR}{dQ} = P \frac{dQ}{dQ} + Q \frac{dP}{dQ} \quad \text{and} \quad MR = P + Q \frac{dP}{dQ}$$

$$P = MR - Q \frac{dP}{dQ} \text{ since } \frac{dP}{dQ} \text{ is negative. } P > MR$$

P is always greater than MR. It is so because the firm has to reduce price for all the units of output to sell additional units of output.

Costs: It is similar to competitive firms. ATC, AVC, and MC curves are U shaped and AFC is rectangular hyperbola. The marginal cost curve is not the supply of the monopolist as in the case of pure competition.

5.2. Short-Run and long-Run Equilibrium

A. Short-Run Equilibrium

The monopolist maximizes his short run profit if the following two conditions are fulfilled.

1. The *marginal cost* is equal to the *marginal revenue*.
2. The slope of marginal cost is *greater than* the slope of the marginal revenue at the point of the intersection.

Proof: The monopolist aims at the maximization of his profit

$$\Pi = R - C$$

(a) *The first – order condition for maximum profit Π*

$$\frac{\partial \Pi}{\partial Q} = 0$$

$$\frac{\partial \Pi}{\partial Q} = \frac{\partial R}{\partial Q} - \frac{\partial C}{\partial Q} = 0 \quad \text{or}$$

$$\frac{\partial R}{\partial Q} = \frac{\partial C}{\partial Q}$$

That is $MR = MC$

The second – order condition for maximum profit

$$\Rightarrow \frac{\partial^2 \Pi}{\partial Q^2} < 0$$

$$\Rightarrow \frac{\partial^2 \Pi}{\partial Q^2} = \frac{\partial^2 R}{\partial Q^2} - \frac{\partial^2 C}{\partial Q^2} < 0$$

or

$$\Rightarrow \frac{\partial^2 R}{\partial Q^2} < \frac{\partial^2 C}{\partial Q^2}$$

$$[\text{Slope of MR}] < [\text{Slope of MC}]$$

A Numerical Example:

Given the demand curve of the monopolist

$$Q = 50 - 0.5P \text{ and } 50 - Q = 0.5P$$

Which may be solved for P,

$$P = 100 - 2Q$$

Given the cost function of the monopolist

$$C = 50 + 40Q$$

The goal of the monopolist is to maximize profit

$$\Pi = R - C$$

(i) We first find the MR

$$TR = QP = Q(100 - 2Q)$$

$$TR = 100Q - 2Q^2$$

$$MR = \frac{\partial R}{\partial Q} = 100 - 4Q$$

(ii) Next find the MC

$$TC = 50 + 40Q$$

$$MC = \frac{\partial TC}{\partial Q} = 40$$

(iii) Equate MR = MC

$$100 - 4Q = 40 \quad 100 - 40 = 4Q$$

$$60 = 4Q \quad Q = 60/4 = 15$$

(iv) The monopolist's price is found by substituting $Q = 15$ into the demand-price equation

$$P = 100 - 2Q = 70$$

(v) The profit is $\Pi = TR - TC$

$$= PQ - TC = 70(15) - [50 + 40(15)]$$

$$= 1050 - 650 = 400$$

This profit is maximum possible, since the second-order condition is satisfied:

(a) When $\frac{\partial^2 R}{\partial Q^2} < \frac{\partial^2 C}{\partial Q^2}$

$$\frac{\partial MC}{\partial Q} = \frac{\partial(40)}{\partial Q} = 0$$

(b) from $\frac{\partial MR}{\partial Q} = \frac{\partial(100-4Q)}{\partial Q} = -4$ we have $\frac{\partial^2 R}{\partial Q} = -4$, Clearly $-4 < 0$.

So we can say output level 15 maximizes profit; and hence it is optimal. Therefore, we can see that the classical condition for equilibrium of equating the marginal revenue and marginal cost remained intact, except that price is different from marginal revenue (unlike the case of perfect competition).

Graphically: The decision for the maximization of the monopolists' profit is the equality of his MC and MR, provided that the marginal cost cuts the marginal revenue curve from below.

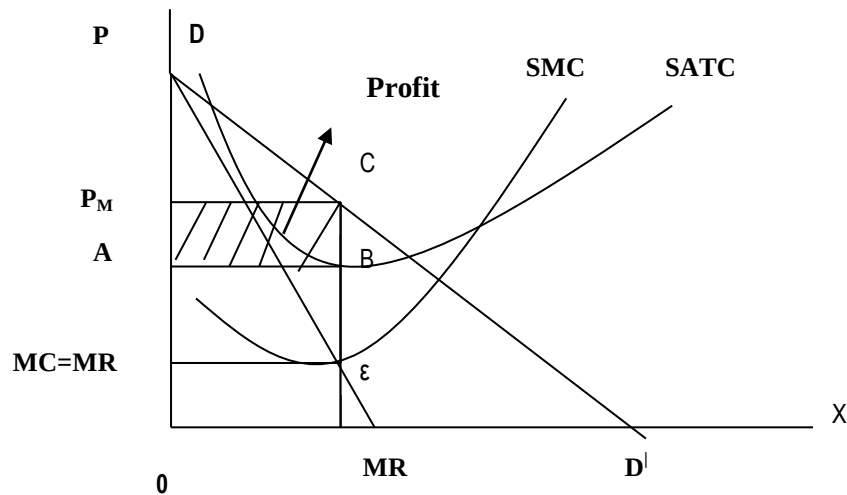


Fig 5.1: Short run Equilibrium of a Monopolist Firm

There is no unique supply curve for the monopolist derived from his MC, the same quantity may be offered at different prices depending on the price elasticity of demand. Graphically:

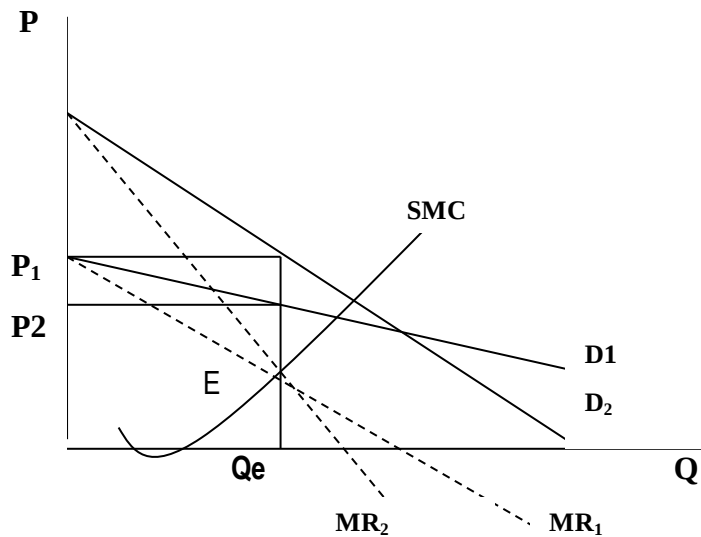


Fig 5.2: Price discrimination of a monopolist
Q_e is sold at P₁ for some consumers and at P₂ for others.

Similarly, given the MC of the monopolist, various quantities may be supplied at any one price, depending on the market demand and the corresponding marginal revenue curve.

Graphically:

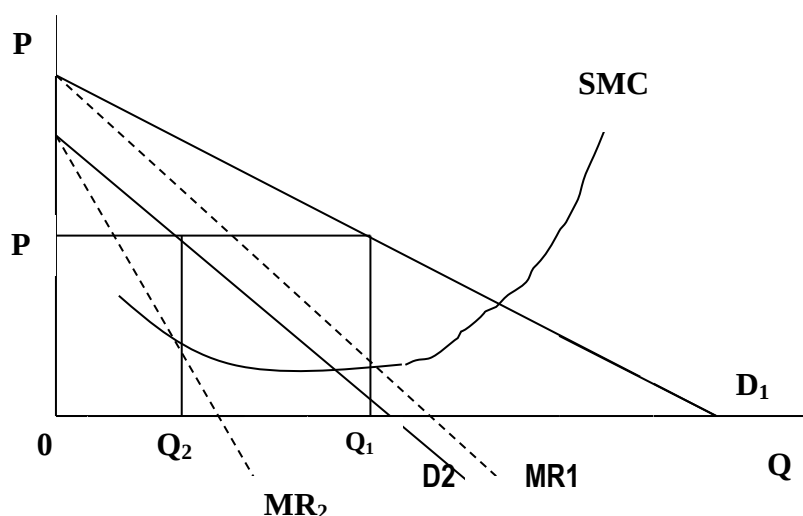


Fig 5.3: Both Q_1 and Q_2 are sold at the same price P .

Since the monopolist can sell the same quantity of output at different prices and can sell different quantities at the same price, depending on elasticity of demand, there is no distinct relationship between price and quantity and supply by a monopolist. To conclude we must say that “the supply curve is not clearly defined.

B. Long run Equilibrium

In the long run the monopolist has the time to expand his plant or to use his existing plant at any level which will maximize his profit. But given entry impossibility into a monopolist market the firm may not necessarily build the optimal plant size. (That is to build up his plant until he reaches the minimum point of the LAC). Given entry barrier, the monopolist may probably continue to earn super normal profits even in the long run.

However, the size of his plant and the degree of utilization of any given plant size depend entirely on the market demand. The monopolist may reach the optimal scale (minimum point of LAC) or remain at suboptimal scale (falling parts of his LAC) or operate beyond the optimal scale (expand beyond the minimum LAC) depending on the market condition.

- (i) For instance, if market is limited/small the firm will maintain a sub-optimal plant and may underutilize the plant.

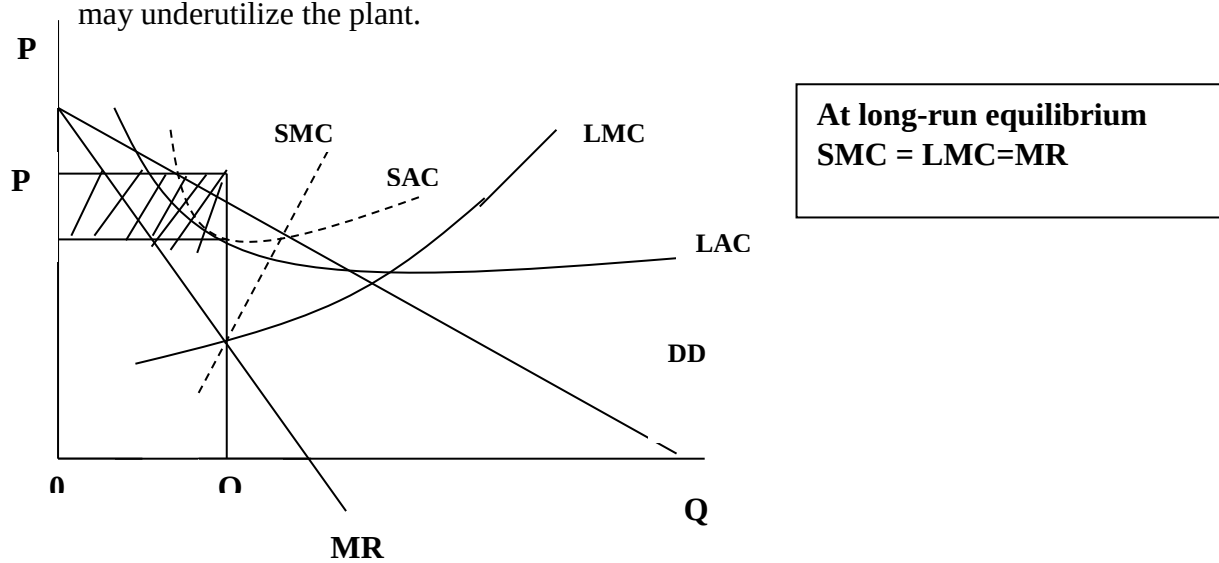


Fig 5.4: Monopolist with suboptimal plant and excess capacity.

The firm is employing plant size smaller than the optimum and it is under utilizing the plant (excess capacity).

- (ii) If the market is so large relative to the expansion path, the firm may build a production plant larger than the optimal and may over-utilize the plant. When the market is larger, the optimum occurs to the right of the minimum point of long run average cost as at point e.

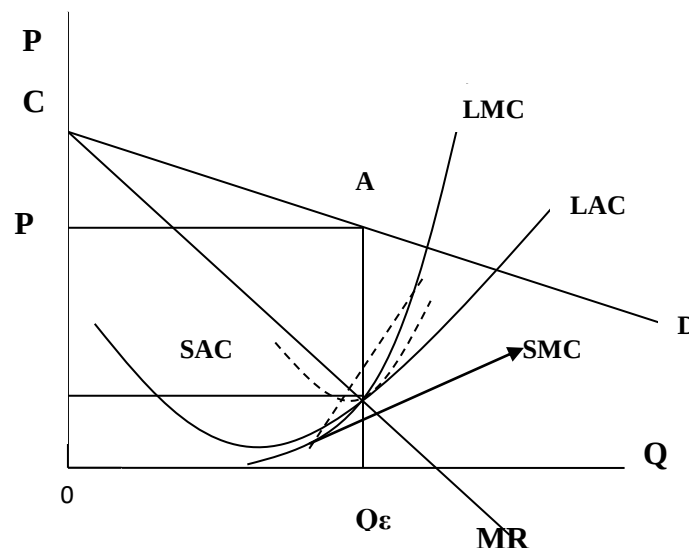


Fig 5.5: Monopolist operating in a large market: his plant is large than the optimal (e) and it is being over-utilized (at ϵ).

(iii) If the market size is just large enough to permit the monopolist to build the optimal plant, the firm will be operating at minimum point of LAC.

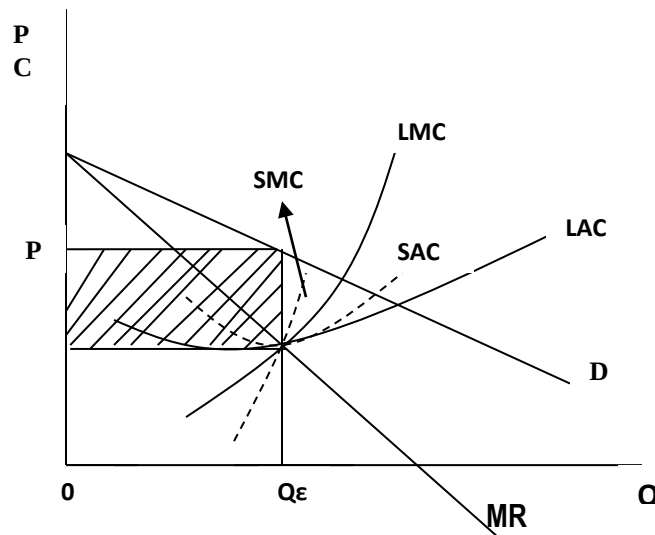


Fig 5.6: Monopolist operating at his optimal plant size (Full capacity utilization)

If the monopolist is at his optimal plant size $SMC = LMC = SAC = MR$ at minimum of LAC.

Note: The firm still earns supernormal profit because price is greater than the marginal revenue (profits in the shaded area).

5.3. Price Discrimination

Price discrimination exists when the same product is sold at different prices to different buyers. The cost of production is either the same or different but not as much as the difference in the changed prices. Ex: different binding of the same book, different locations of seats in a theater etc.,) The necessary conditions for the firm to practice price discrimination are as follows:

1. The monopolist must distinguish between more price elastic consumer and less price elastic consumer. (The market must be divided into sub markets with different price elasticities.)
2. There must be effective separation of the sub markets, (no reselling can take place from a low price market to a high price market). Eg., Services like transport, a show, services of doctor, electricity supply etc.,

By practicing price discrimination, the monopolist can increase revenue and hence profit by selling the optimal output (where $MR = MC$) at different prices in different markets. Monopolists normally practice three kinds price discrimination. First Degree price discrimination: Second Degree Price Discrimination and finally and Third-Degree Price Discrimination.

Third Degree Price Discrimination

Revenue earned from price discrimination is greater than the revenue from a single (uniform) price sale. Since the cost is the same, profit is also higher under price discrimination. This is third degree price discrimination. In this type of price discrimination in which output is sold for two different categories of individuals at different prices but every individual in the same group takes the same price is called third degree price discrimination.

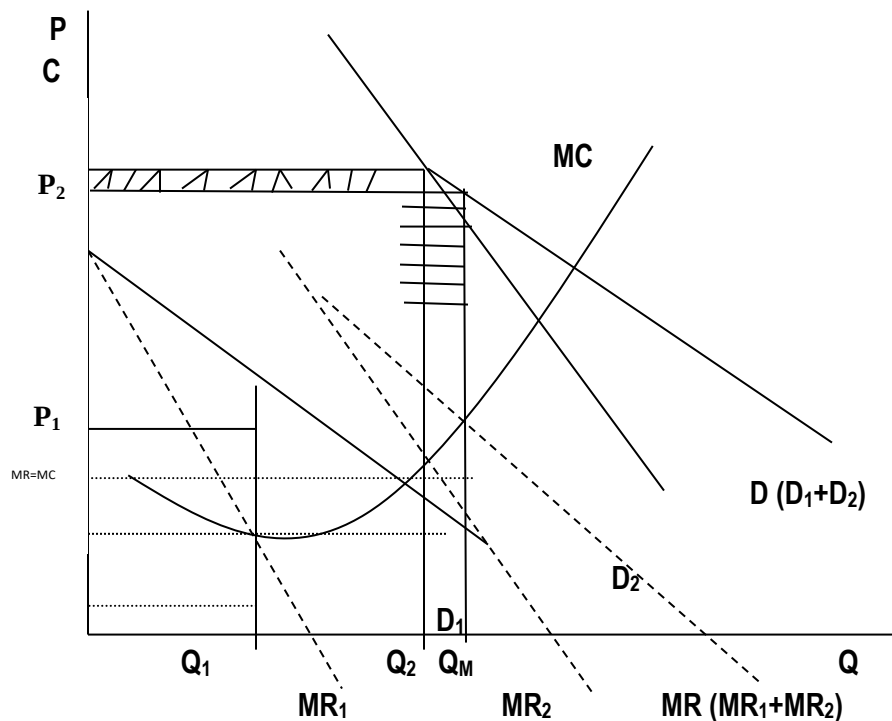


Fig 5.7: Price Discrimination

Suppose the monopolist takes his market into two sub markets (the above figure) based on different elasticity of demand existence. Market 1 with more elastic demand will be treated with demand curve D_1 .

Market 2 with less elastic demand will be treated with demand curve D_2 .

The price discriminating firm has to decide a) The total output that it must produce and b) How much to sell in each market and at what price, so as to maximize its profits? Total quantity to be produced is defined by the point of intersection of the MC and the aggregate MR curves.

If the firm charges uniform price in two markets the total revenue is OPA_Q . But the firm increases its revenue or profits by charging different prices according to $MR_1=MR_2=MC$.

$MR_1=MR_2=MC$ condition is fulfilled at points at which the horizontal dotted line through e ($MR=MC$) intersects MR_1 & MR_2 .

CASE 1: The firm charges uniform (identical price) P where $MC = MR$.

CASE 2: The firm charges different prices depending on the elasticity of demand.

$$TR_1 = OP_1FQ_1$$

$$TR_2 = OP_2EQ_2$$

$$TR = OP_1FQ_1 + OP_2EQ_2 \quad \text{since } OP_1FQ_1 = CBQQ_2$$

$$TR = OP_2EQ_2 + CBQQ_2$$

By practicing price discrimination, the firm gains PP_2ED and losses $ABCD$. But the gain (PP_2ED) is greater than the loss. Revenue from price discrimination is greater than revenue from a uniform price. Since the cost is the same, profit is also higher under price discrimination. Third degree price discrimination can also be illustrated by using two different prices for different quantities on the same market demand curve.

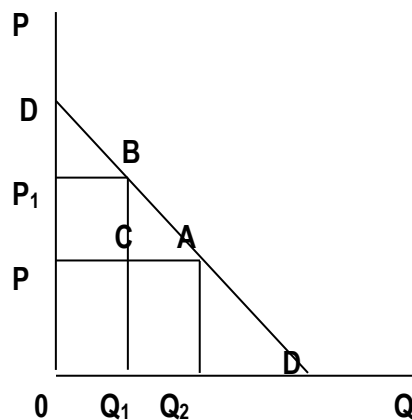


Fig 5.8: Third Degree Price Discrimination

If the total output Q is sold at a uniform price P , total revenue will be $R = OPAQ$. If the firm sells the first OQ_1 units at P_1 and Q_1Q units at P , the total revenue will be greater by P_1BC . At uniform price P , $PDBA$ was consumer surplus. Because of price discrimination the additional revenue is obtained by extracting the part of the consumer surplus (PP_1BC).

Second Degree Price Discrimination:

If the firm negotiates at more than two prices more of the consumers' surplus can be extracted and revenue can be further increased. This way of increasing revenue is called second degree price discrimination. By practicing price discrimination, the firm's revenue has increased by $PP1Ah+ehiB+efcg$, which was a part of consumers' surplus.

First Degree Price Discrimination:

In the extreme case, the firm might negotiate price with every consumer to change his/her the maximum price. He/she is willing to pay (reservation price). It extracts the entire consumer's surplus and it is called first degree price discrimination. In this case the MR is the same as the price. The contribution of every unit of output sold at its reservation price is MR.

Exercise:

Consider a monopolist that takes a market which can be divided into two submarkets with demand functions.

$$Q_1 = 100 - P_1 \quad Q = Q_1 + Q_2$$

$$Q_2 = 100 - 2P_2 \quad = 200 - 3P$$

If MC is constant at \$10 calculate the prices that must be charged in the two markets and quantities if the firm practices price discrimination.

Solution:

First change the direct demand functions in to inverse demand functions.

$$Q_1 = 100 - P_1 \quad Q_2 = 100 - 2P_2$$

$$P_1 = 100 - Q_1 \quad P_2 = 50 - \frac{1}{2} Q_2$$

$$R_1 = P_1 Q_1 \quad R_2 = P_2 Q_2$$

$$Q_1[100 - Q_1] \quad Q_2[50 - \frac{1}{2} Q_2]$$

$$100Q_1 - Q_1^2 \quad 50Q_2 - \frac{1}{2} Q_2^2$$

$$MR_1 = 100 - 2Q_1 \quad MR_2 = 50 - Q_2$$

Set MR in each market with MC

$$MR_1 = MR_2 = MC$$

$$100 - 2Q_1 = 10 \quad 50 - Q_2 = 10$$

$$100 - 10 = 2Q_1 \quad 50 - 10 = Q_2$$

$$Q_1 = 90/2 = 45 \quad 40 = Q_2$$

$$Q_1 = 100 - P_1 = 45$$

$$Q_2 = 100 - 2P_2 = 40$$

$$P_1 = 100 - 45 = 55$$

$$= 100 - 40 = 2P_2$$

$$P_2 = 60/2 = 30$$

5.4. Multi plant Monopolist

A monopolist can produce identical product in different plants. For simplicity we assume the firm has only two plants, Plant A and Plant B, each with different cost structures. In this case the monopolist has to make two decisions.

- Price and output levels.
- How much to produce in the first plant and how much in the second plant.

The monopolist should know the market demand curve the corresponding MR curve and cost structure of those plants.

$$MC = MC_1 + MC_2$$

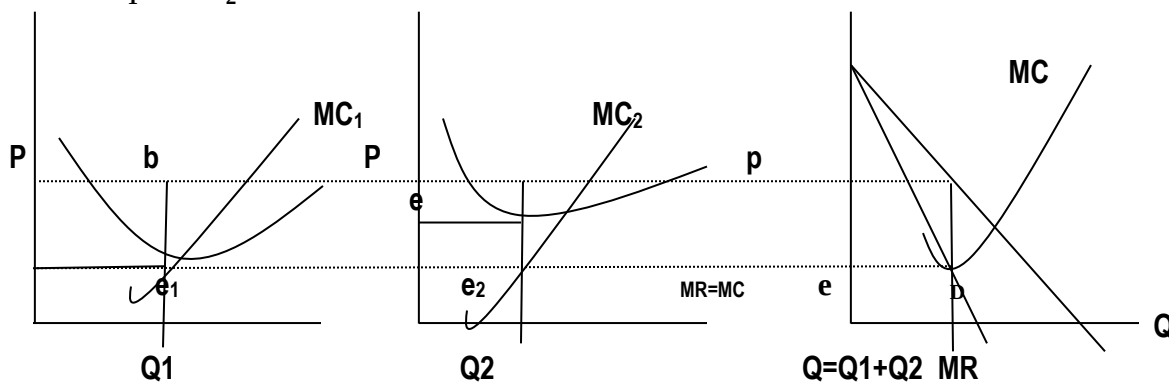


Fig 5.9: Multi Plant monopolist with same Price

Optimum point is obtained by the equality point of MC_1 , MC_2 and MR .

In other words, the monopolist maximizes his profit by utilizing each plant up to the level at which the marginal costs are equal to each other and to the common marginal revenue. This is because if the MC on one plant, say plant A, is lower than the marginal cost of plant B, the monopolist would increase his profits by increasing the production in A and decreasing it in B, until the condition $MC_1 = MC_2 = MR$ is fulfilled.

Graphically the equilibrium of the multi-plant monopolist may be defined as follows. The total profit-maximizing output and its price is defined by the intersection of MC and MR curves Point E in the diagram. From the point of intersection we draw a line, parallel to the X-axis, until it intersects the individual MC_1 and MC_2 curves of the two plants. At these points the equilibrium

condition ($MC = MR = MC_1 = MC_2$) is satisfied. If from these points (E1 and E2) we draw perpendiculars to the X-axis of plant A and plant B, We find the level of output that will be produced in each plant. Clearly $Q_1 + Q_2$ must be equal to the profit maximizing output Q. The total profit is the sum of profits from products of the two plants. The profit from plant A is the shaded area **abcd** and the profit from plant B is the shaded **efih**.

Numerical Example: Given the monopolist's cost and demand curve.

$$Q = 200 - 2P \quad \text{or} \quad P = 100 - 0.5Q$$

$$C_1 = 10X_1 \quad C_2 = 0.25X_2^2.$$

Find Q, Q_1 , and Q_2 , P Where $Q = Q_1 + Q_2$.

$$P = 100 - 0.5Q$$

$$TR = P \cdot Q = 100Q - 0.5Q^2$$

$$\begin{aligned} MR &= 100 - Q \quad Q = Q_1 + Q_2 \\ &= 100 - Q_1 - Q_2 \end{aligned}$$

$$TC_1 = 10X_1$$

$$MC_1 = 10$$

$$TC_2 = 0.25 Q_2^2$$

$$MC_2 = 0.5Q_2$$

$$MC_1 = MR \quad 10 = 100 - Q_1 - Q_2$$

$$MC_2 = MR \quad 0.5Q_2 = 100 - Q_1 - Q_2$$

$$Q_1 + Q_2 = 90$$

$$1.5Q_2 + Q_1 = 100$$

$$-Q_1 - Q_2 = -90$$

$$1.5Q_2 + Q_1 = 100$$

$$Q_1 + Q_2 = 90$$

$$0.5Q_2 = 10 \quad Q_1 = 90 - 20 = 70$$

$$Q_2 = 20$$

$$P = 100 - 0.5(90) = 100 - 45 = 55$$

$$\text{Monopolist profit } \Pi = R - C_1 - C_2$$

$$= 4950 - 10(20) - 0.25(4,900) = 4150$$

5.5. Social Costs Of Monopoly power

Price is higher and output is lower in the monopoly market model than the competitive market model. Because of these monopoly is said to be socially inefficient in allocating factors of production. In Competitive Market $P = MC$

Monopoly Market $P > MC$

To calculate the degree of inefficiency due to monopoly we can use the concepts of producer's surplus and consumer's surplus.

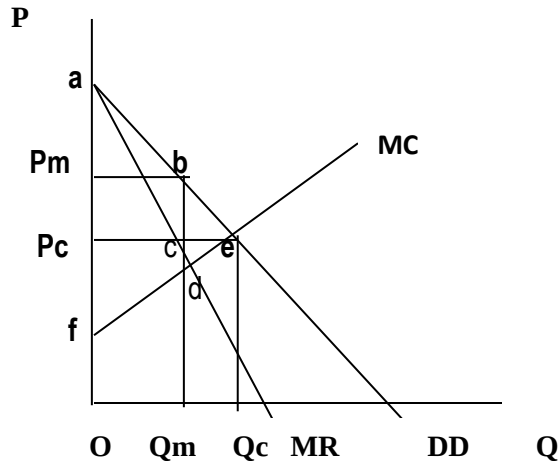


Fig 5.10: Producer and consumer Surplus

Under Competition the net welfare gained by the consumers and producers is as follows:

$$\text{Net Welfare} = \underbrace{apce}_{\text{Consumer's Surplus}} + \underbrace{Pcef}_{\text{Producer's Surplus}}$$

Under Monopoly the net welfare gained by the consumers in $aPmb$ which is less than net welfare gained under competitive ($aePc$). $PmbcPc$ is taken by producers. But the area bed is totally lost. (Which is not consumer surplus & Producers surplus). Area bed is dead weight loss due to monopoly allocation as opposed to competitive market.

Chapter –Five Review Questions

Part I: Read each expression and choose the most appropriate answer.

- The best level of output for the pure monopolist occurs at the point where
 - TC is minimum
 - $TR = TC$

- C. $MR = MC$
 - D. TR and TC curves are parallel
2. Price discrimination will always lead to
 - A. Increase in output
 - B. Decrease in output
 - C. no change in output
 - D. Increase in TR
 3. Discriminating monopoly is possible if two markets have
 - A. Rising cost curves
 - B. Rising and declining cost curves
 - C. Different elasticities of demand
 - D. Equal elasticities of demand
 4. Which of the following statements is not true? A discriminating monopolist
 - A. Operates in more than one market
 - B. Makes more profit because he discriminates
 - C. Maximizes his profits in each market
 - D. Charges same prices in different markets
 5. If a profit maximizing monopoly's MR exceeds its MC, then it should
 - A. Raise its price
 - B. Rise its output
 - C. lower its output
 - D. lower its price
 6. One of the following characterizes monopoly market structure:
 - A. Only a single seller of a product that have large number of buyers
 - B. Market price set by a producer
 - C. Barrier to entry
 - D. All of the above

Part Two: Work Out and Discussion Questions.

1. Suppose a monopolist faces a market demand curve given by $P = 55 - 2Q$ and its total costs of production given by $TC = 100 - 5Q + Q^2$
 - a) Determine the profit maximizing price and output.
 - b) Calculate the dead weight loss from the monopoly power.
2. Explain and show graphically the multi plant monopolist marketing situation.